**1. O/N/23/22/Q8**

Fig. 8.1 shows a permanent magnet attached to one end of an unstretched spring and held at rest just above a solenoid.

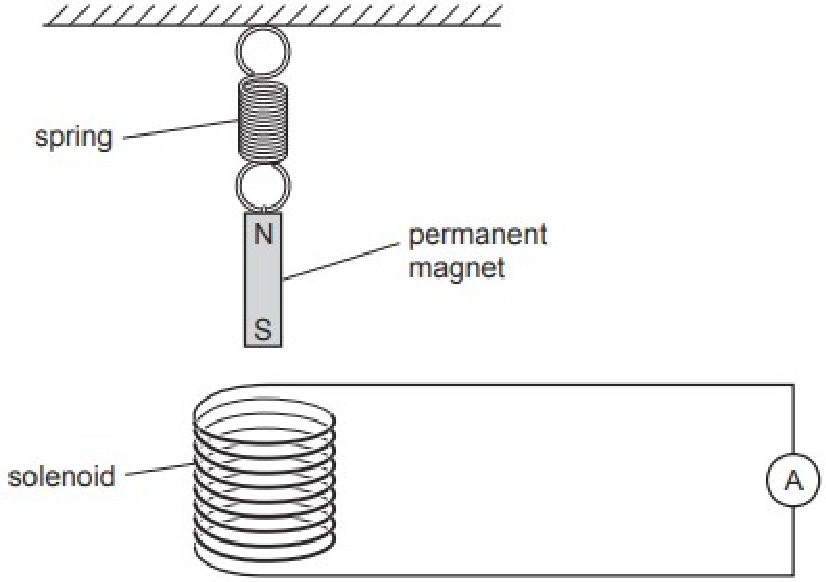


**Fig. 8.1**

The solenoid is connected to a sensitive ammeter.

**(a)** State the name of a substance from which permanent magnets are made.

............................................................................................................................................ [1]

**(b)** The magnet is released and moves downwards towards the solenoid.

The ammeter shows that there is a current in the solenoid when the magnet moves.

**(i)** Explain why there is a current in the solenoid.

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............................................................................................................................................ [3]

**(ii)** The current causes the top end of the solenoid to become the S pole of an electromagnet.

The bottom end of the permanent magnet is an S pole.

Explain why the top end of the solenoid becomes an S pole as the magnet moves downwards towards the solenoid.

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.................................................................................................................................... [2]

**(c)** The magnet oscillates up and down.

Describe what happens to the current in the solenoid.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

[Total: 8]

**2. O/N/23/21/Q7**

An alternating current (a.c.) generator is used in a remote location to supply electricity.

**(a)** Fig. 7.1 shows the structure of a simple a.c. generator with the coil horizontal.

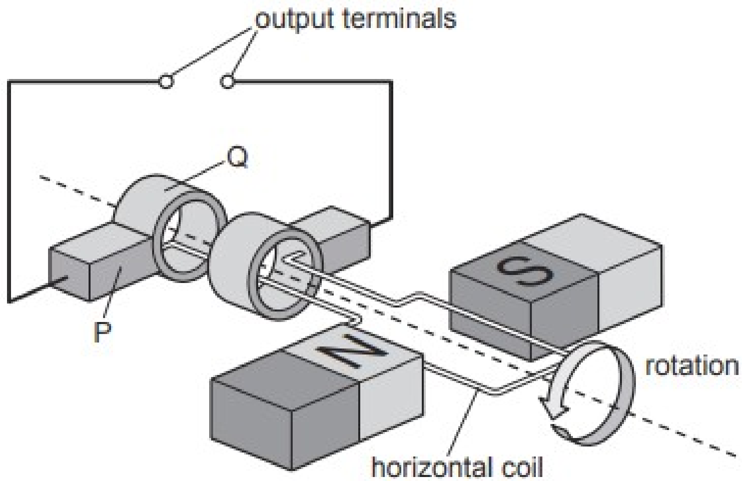


**Fig. 7.1**

**(i)** State the name of the components labelled P and Q on Fig. 7.1.

P ..........................................................................................................................

Q .......................................................................................................................... [1]

**(ii)** When the generator is producing an electrical output, P is stationary and Q is rotating.

Explain the purpose of P and Q.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(b)** The output of the generator is connected to an oscilloscope.

On the oscilloscope the Y-gain is set at 5.0 V/cm and the timebase setting is 2.0 ms/cm.

Fig. 7.2 shows the trace on the screen of the oscilloscope.

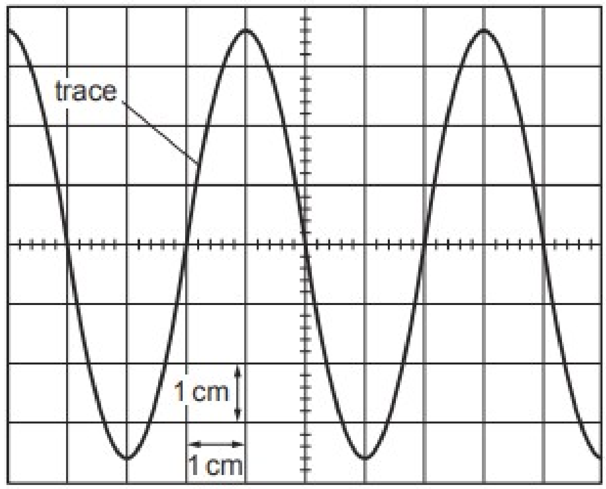


**Fig. 7.2**

**(i)** Write a letter X on the trace shown in Fig. 7.2 to indicate one point when the coil is vertical. [1]

**(ii)** Using Fig. 7.2, determine the maximum value of the e.m.f. produced by the generator.

maximum e.m.f. = ........................................................ V [2]

**(iii)** Using Fig. 7.2, determine, in milliseconds, the time it takes for one revolution of the coil.

time for one revolution of the coil = ..................................................... ms [1]

**(iv)** Determine the frequency of the output of the generator.

frequency = ...................................................... Hz [2]

[Total: 9]

**3. M/J/23/22/Q8**

Fig. 8.1 shows a step-down transformer used to operate an electric bell.

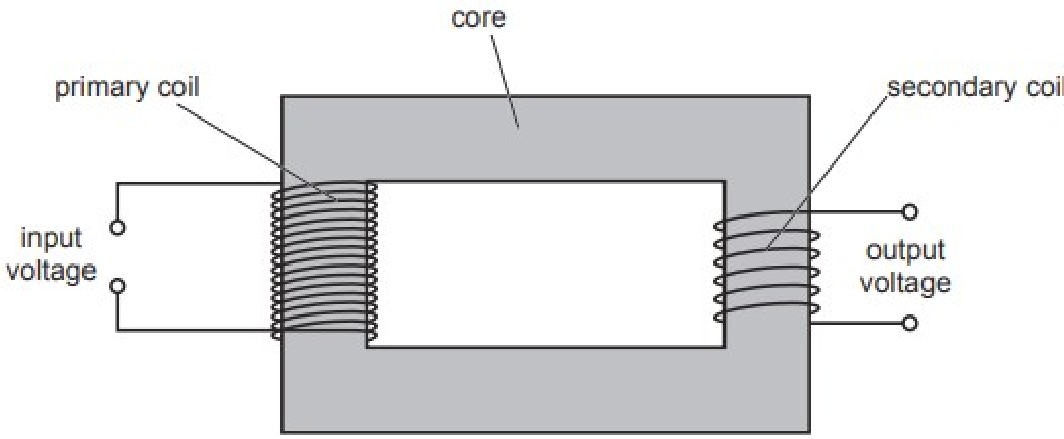


**Fig. 8.1**

**(a)** State the material used for the core of the transformer.

.......................................................................................................................... [1]

**(b)** A current in the primary coil produces a magnetic field in the core.

Explain how an alternating voltage is produced in the secondary coil.

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..........................................................................................................................

.......................................................................................................................... [3]

**(c)** The transformer has 4600 turns on the primary coil which is connected to a mains supply of 230 V.

An output of 5.0 V is used to operate the bell.

Calculate the number of turns needed on the secondary coil.

number of turns = ........................................................ [2]

**(d)** State one change that can be made to the transformer shown in Fig. 8.1 so that it can be used as a step-up transformer.

..........................................................................................................................

.......................................................................................................................... [1]

**4. M/J/23/21/Q9**

Fig. 9.1 shows a simple a.c. generator.

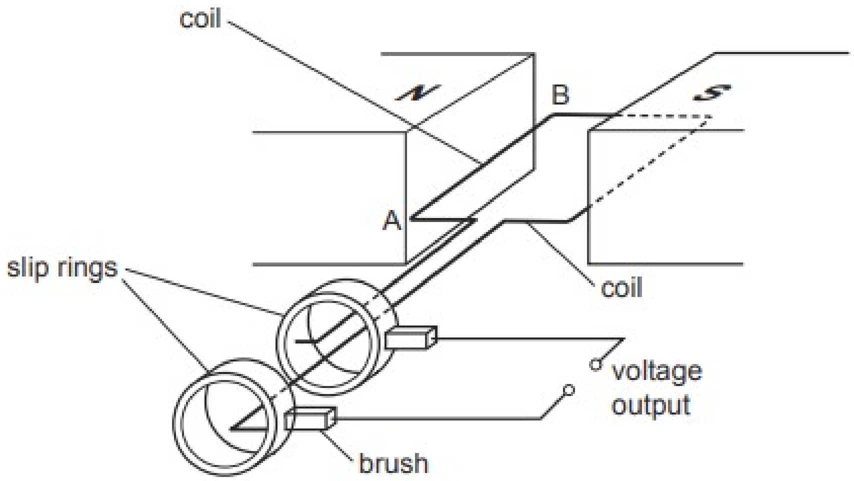


**Fig. 9.1**

**(a)** Explain why there is a voltage induced in the coil when the coil is turned.

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..........................................................................................................................................

..........................................................................................................................................

...................................................................................................................................... [2]

**(b)** In Fig. 9.1 the coil is horizontal, with side AB on the left. The output voltage is +6.0V.

On Fig. 9.2 draw a line from each of the shaded boxes to one of the circled voltages to show the voltage output when the coil is in different positions.

One line has been drawn for you.

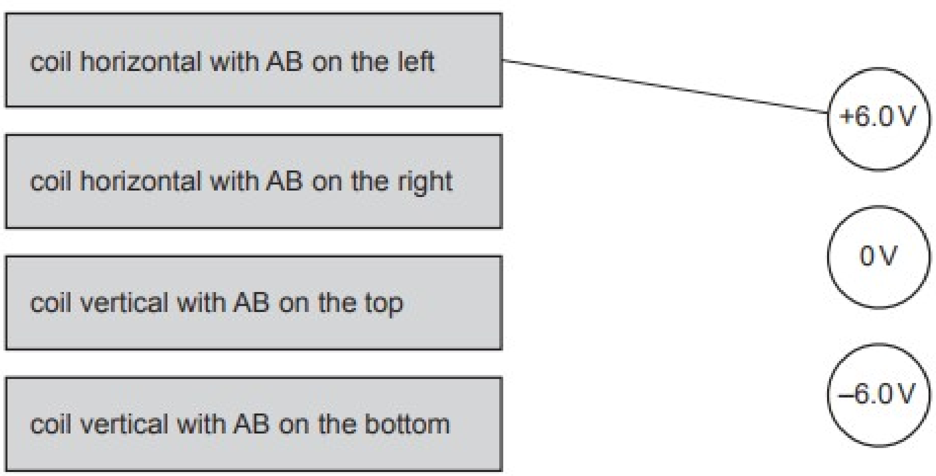


**Fig. 9.2**

[2]

**(c)** Both an a.c. generator and a d.c. motor contain a coil and brushes.

**(i)** Fig. 9.1 shows how the brushes are connected to the coil in an a.c. generator.

Draw a diagram to show how the brushes are connected to the coil in a d.c. motor.

[2]

**(ii)** State why there are forces on the sides of the coil in a d.c. motor.

..........................................................................................................................................

.................................................................................................................................. [1]

[Total: 7]

**5. O/N/22/22/Q6**

The primary coil of a transformer is connected to the mains supply. The voltage of the a.c. mains supply is 240 V.

Fig. 6.1 is a diagram of the arrangement.

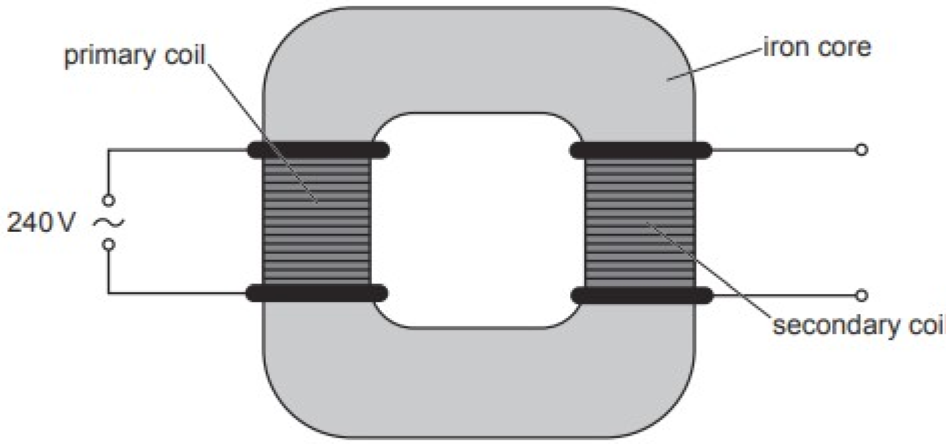


**Fig. 6.1**

**(a)** Explain why a voltage is produced in the secondary coil.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [3]

**(b)** There are 5600 turns on the primary coil of the transformer and 350 turns on the secondary coil.

**(i)** Calculate the output voltage of the transformer.

output voltage = ........................................................ [2]

**(ii)** The output of the transformer is connected to a 90 W filament lamp which operates at normal brightness.

Calculate the current in the lamp.

current = ........................................................ [2]

**6. O/N/22/21/Q6**

An electromagnet is used to separate objects that are magnetic from objects that are non-magnetic.

Fig. 6.1 shows the electromagnet suspended from the arm of a crane.

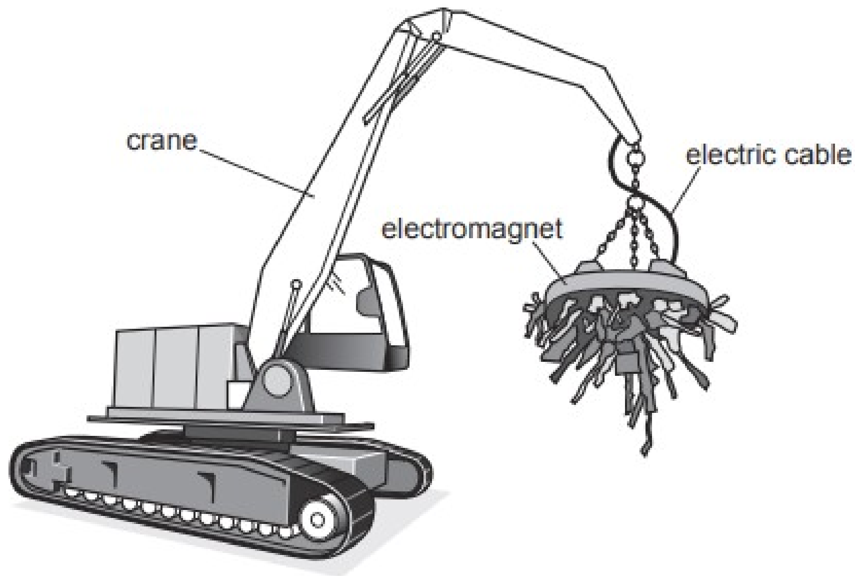


**Fig. 6.1**

**(a)** State the name of the metal from which the core of the electromagnet is made.

............................................................................................................................ [1]

**(b)** The electromagnet is powered by a 220 V d.c. (direct current) supply.

The electromagnet is switched on and the current in the circuit is 39 A.

**(i)** Calculate the power transferred to the electromagnet.

power = ........................................................... [2]

**(ii)** The d.c. supply is a set of batteries.

The batteries power the magnet for 4.5 hours before they need to be replaced.

Calculate the charge driven around the complete circuit in this time.

charge = ........................................................ [3]

**(c)** On a very cold morning, when the electromagnet is first switched on, the current is greater than 39A before decreasing to the usual value.

Explain why the current is initially greater than 39A.

..........................................................................................................................................

........................................................................................................................................ [1]

[Total: 7]

**7. O/N/22/21/Q9**

The power supply in an electric circuit is a battery of electromotive force (e.m.f.) 12 V.

**(a)** State **two** ways in which the e.m.f. of a battery differs from that of an alternating current (a.c.) power supply.

1 ..........................................................................................................................................

2 ..........................................................................................................................................

[2]

**(b)** The circuit includes three resistors and two open switches, $S_1$ and $S_2$.

Fig. 9.1 shows the circuit.

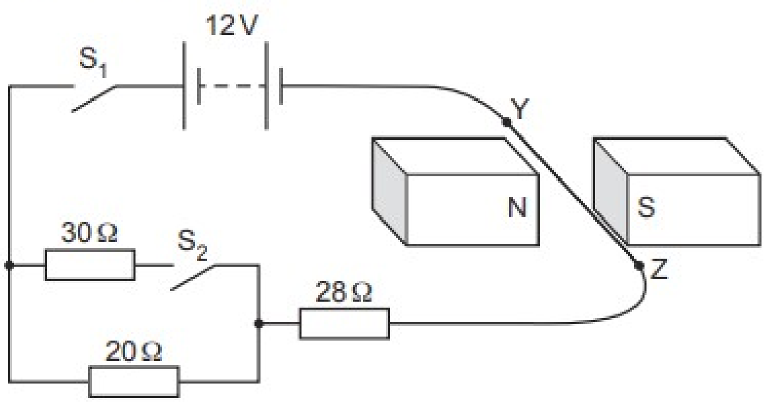


**Fig. 9.1**

YZ is a straight, horizontal section of connecting wire that lies between two magnets.

$S_1$ is now closed.

**(i)** Calculate the current in YZ.

current = ............................................................ [2]

**(ii)** Explain why YZ experiences a force.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**(iii)** Tick the box which describes the direction of the force on YZ.

| | | | |
|---|---|---|---|
| towards N | ☐ | towards Z | ☐ |
| towards S | ☐ | downwards | ☐ |
| towards Y | ☐ | upwards | ☐ |

[1]

**(iv)** Explain how the direction of the force on YZ is determined.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(c)** Switch $S_2$ in the circuit in Fig. 9.1 is now also closed.

**(i)** Calculate the total resistance of the circuit.

resistance = ........................................................ [3]

**(ii)** Explain what happens to the force on YZ as switch $S_2$ is closed.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(iii)** The current in the 20 Ω resistor is $I_{20}$. The current in the 30 Ω resistor is $I_{30}$.

State a value for the ratio $I_{20}/I_{30}$.

ratio $I_{20}/I_{30}$ = ........................................................ [1]

**8. M/J/22/22/Q6**

**(a)** Fig. 6.1 shows part of a toy which contains two ring-shaped, permanent magnets. A plastic rod passes through the centre of both magnets.

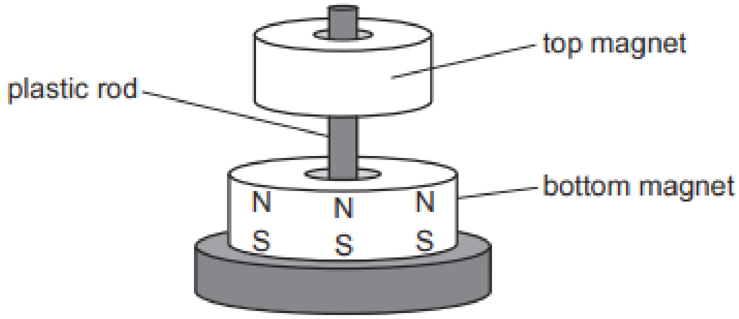


**Fig. 6.1**

The top magnet can move up and down freely around the plastic rod.

The magnetic poles on the bottom magnet are shown in Fig. 6.1.

**(i)** The top magnet floats in the air above the bottom magnet.

On Fig. 6.1, mark the poles on the top magnet and explain why it floats in the air above the bottom magnet.

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....................................................................................................................................... [2]

**(ii)** The top magnet is replaced with a ring made of iron.

Explain why the iron ring sticks to the bottom magnet.

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....................................................................................................................................... [2]

**(b)** A wire carrying a current passes at right angles through a piece of paper.

Fig. 6.2 shows a cross and circle where the current in the wire passes into the plane of the paper.

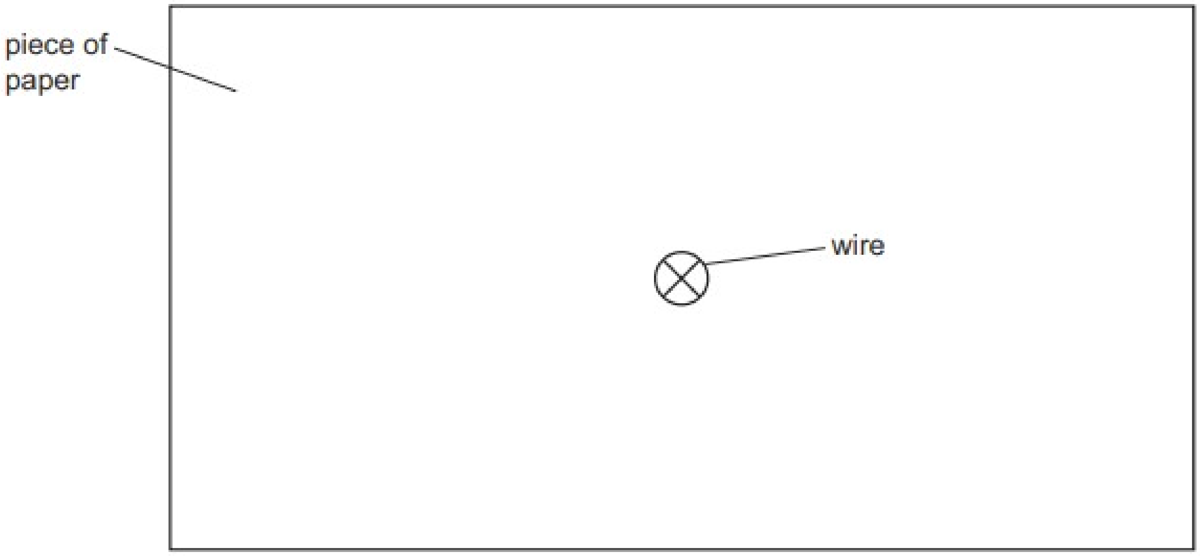


**Fig. 6.2**

On Fig. 6.2, sketch **three** magnetic field lines to show the magnetic field pattern around the wire.

Show the direction of the field on your sketch. [2]

[Total: 6]

**9. M/J/22/21/Q5**

Fig. 5.1 shows a simple d.c. motor used in a toy car.

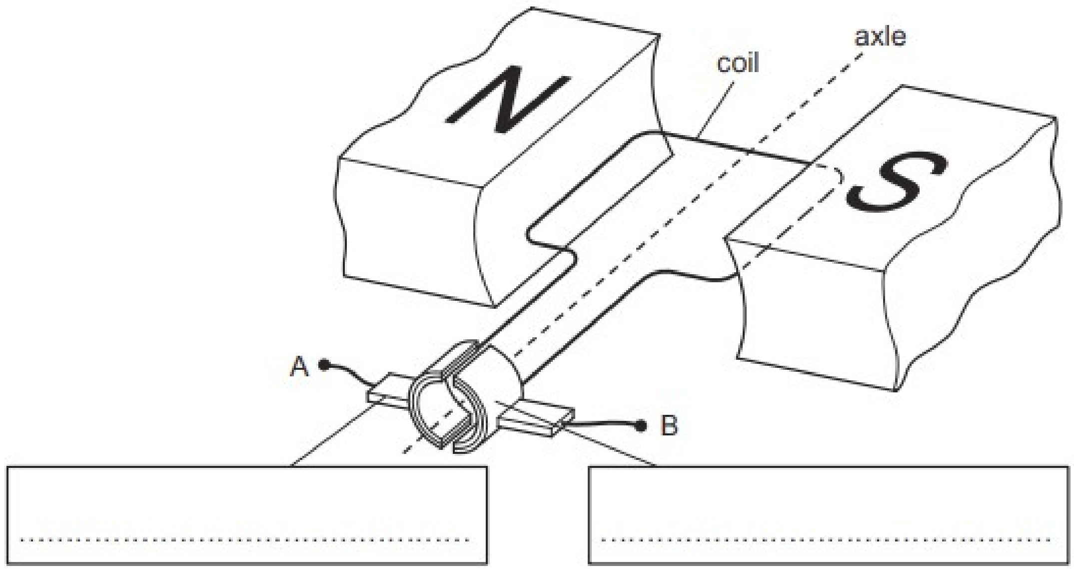


**Fig. 5.1**

Terminals A and B are connected to a battery and the motor rotates.

**(a)** On Fig. 5.1, add labels inside the boxes to identify the parts of the motor indicated. [2]

**(b)** State what happens to the rotation of the coil when:

**(i)** the number of turns on the coil is increased

........................................................................................................................ [1]

**(ii)** the magnetic field between the poles of the magnet is reversed.

........................................................................................................................ [1]

**(c)** The power supply to the motor is switched on and off at a steady rate.

Fig. 5.2 shows how the speed of the toy car varies with time as a result of the power supply being switched on and off.

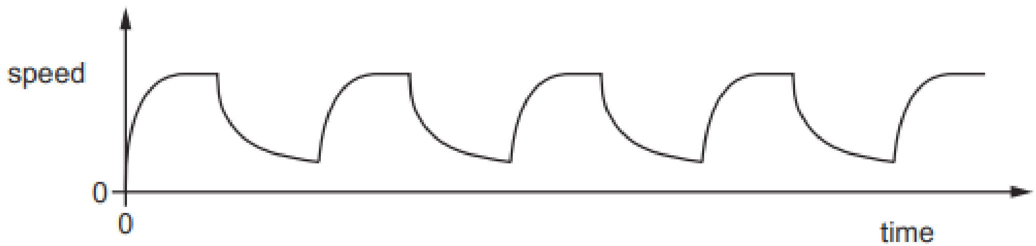


**Fig. 5.2**

**(i)** Describe the motion of the toy car.

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..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(ii)** The voltage supplied to the motor is switched on for a longer period of time and off for a longer period of time, at a steady rate.

Suggest what happens to the motion of the toy car.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [1]

[Total: 7]

**10. O/N/21/21/Q7**

Fig. 7.1 shows a solenoid connected to an ammeter.

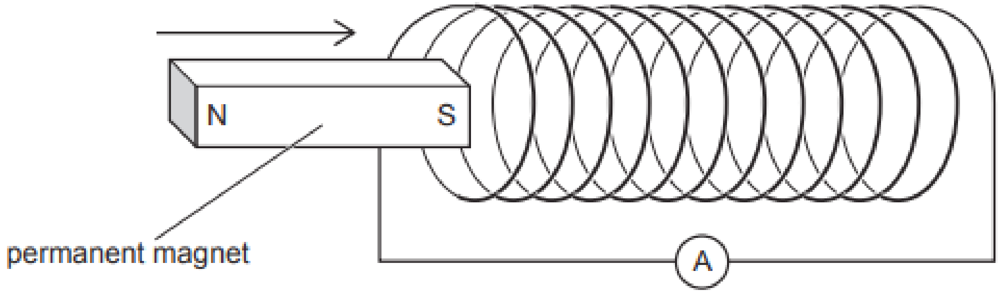


**Fig. 7.1**

The S-pole of a permanent magnet moves into the left-hand end of the solenoid.

The ammeter reading shows that there is a small positive current in the circuit.

**(a)** Explain why there is a current in the circuit when the magnet moves.

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..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [3]

**(b)** When the magnet is inside the solenoid, it stops moving. It is then pulled back out of the solenoid.

Explain what happens to the ammeter reading as the magnet moves out of the left-hand end of the solenoid.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**[Total: 5]**

**11. M/J/21/22/Q7**

Fig. 7.1 shows part of a simple d.c. electric motor.

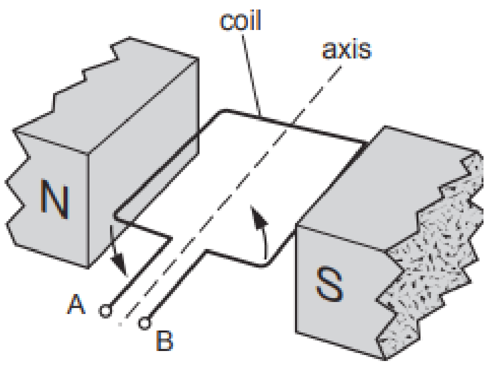


**Fig. 7.1**

The motor is connected to a battery with the positive terminal of the battery connected to terminal A.

**(a)** Explain why the coil turns in the direction shown.

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.......................................................................................................................................... [3]

**(b)** The turning effect is increased when the coil is wound around a soft-iron cylinder.

**(i)** Explain why this happens.

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..........................................................................................................................................

.......................................................................................................................................... [2]

**(ii)** Suggest **one** other way to increase the turning effect of the motor.

.......................................................................................................................................... [1]

[Total: 6]

**12. M/J/21/21/Q6**

A student moves a metal bar upwards between the poles of a magnet, as shown in Fig. 6.1.

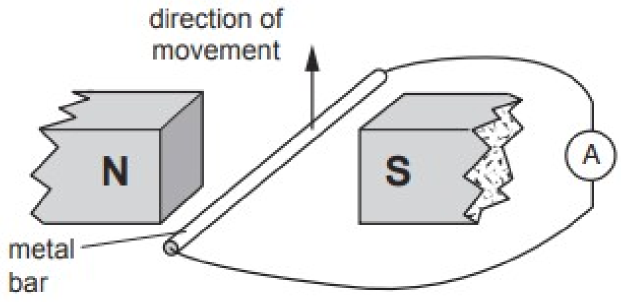



**Fig. 6.1**

The ammeter connected to the metal bar shows a small positive reading as the bar moves.

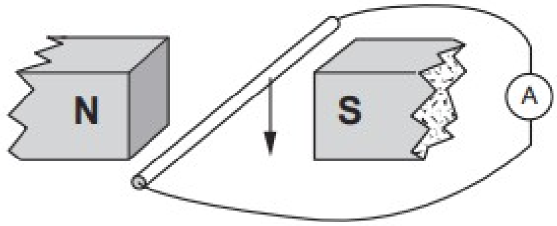
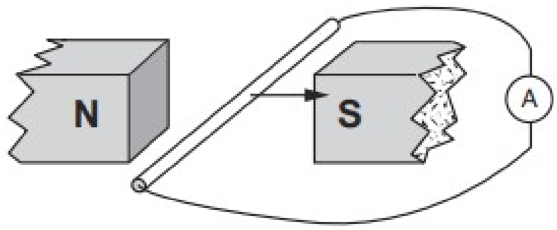
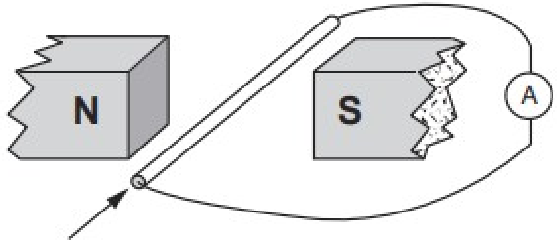
**(a)** Explain why there is a reading on the ammeter.

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** Complete Table 6.1 by stating what the ammeter shows when the metal bar moves in the direction shown by the arrow in each diagram.

**Table 6.1**

| | reading shown on the ammeter |
|---|---|
|  | ............................................................ ............................................................ |
|  | ............................................................ ............................................................ |
|  | ............................................................ ............................................................ |

[3]

**(c)** The student finds that using a stronger magnet increases the reading on the ammeter.

State **one** other way in which the student can produce a larger reading on the ammeter, using the same rod.

..........................................................................................................................................

.......................................................................................................................................... [1]

**(d)** Describe how Lenz's law applies when the bar is moved upwards.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [1]

[Total: 7]

**13. O/N/20/22/Q9**

Fig. 9.1 shows a permanent magnet lying on a piece of paper.

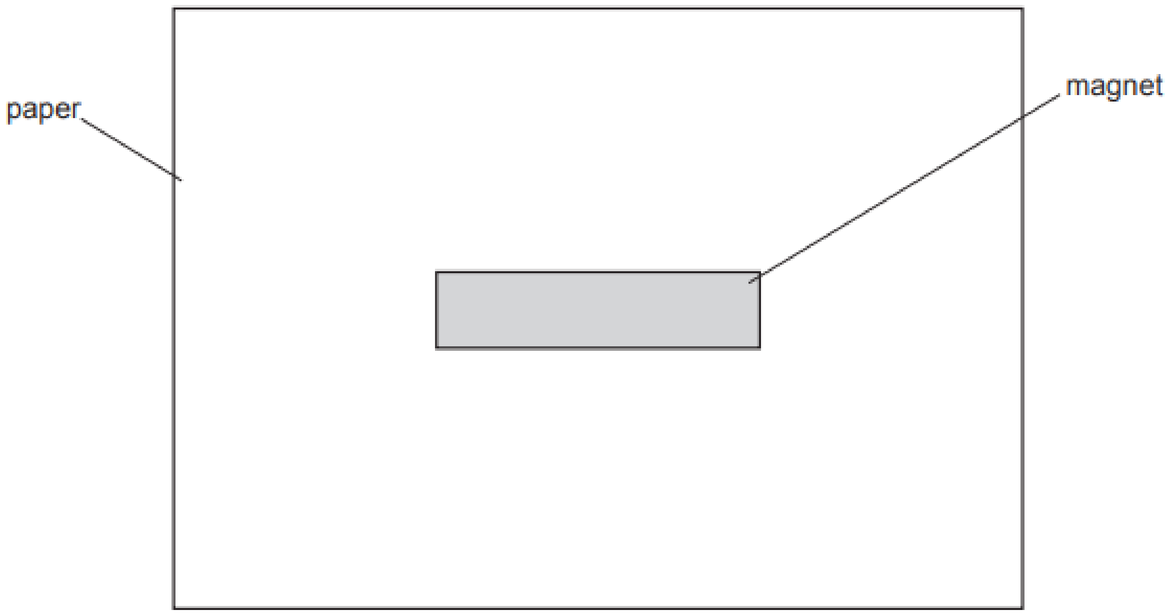



**Fig. 9.1**

**(a)** Underline the material in the list from which it is possible to make a strong, permanent magnet.

**aluminium** **copper** **glass** **iron** **mercury** **plastic** **steel**

[1]

**(b)** Describe an experiment to plot the pattern and the direction of the magnetic field surrounding the magnet. You may draw on Fig. 9.1, if you wish.

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.................................................................................................................................... [4]

**(c)** Fig. 9.2 shows the N-pole of a magnet placed in front of the S-pole of a second magnet.

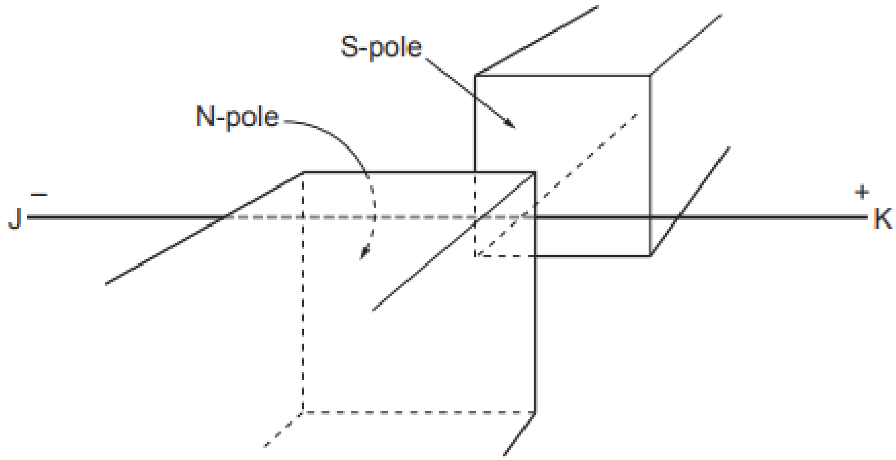


**Fig. 9.2**

A section of a horizontal, metal wire JK lies in the magnetic field between the two magnetic poles. End K of the metal wire is connected to the positive terminal of a battery and end J is connected to the negative terminal.

**(i)** Explain in terms of electrons, why there is a current in the wire and state the direction of the conventional current.

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..........................................................................................................................................

.................................................................................................................................... [2]

**(ii)** The part of JK that is in the magnetic field experiences a force F.

State the direction of F and describe how this direction is deduced.

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..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [4]

**(iii)** The equipment in Fig. 9.2 is used in a similar experiment.

The part of JK that lies between the poles of the magnets, now passes through a long iron tube that is fixed in position.

The tube is shown in Fig. 9.3.

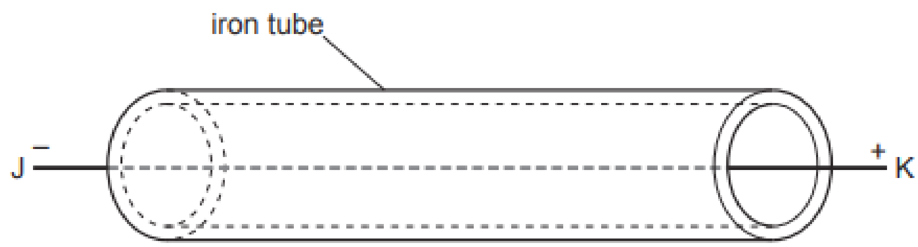


**Fig. 9.3**

JK is connected to the battery in the same way as before.

State what happens to force F on the wire.

Explain your answer.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(d)** The iron tube and the wire JK are removed.

A square, vertical coil is placed between the poles so that the plane of the coil lies in the magnetic field as shown in Fig. 9.4.

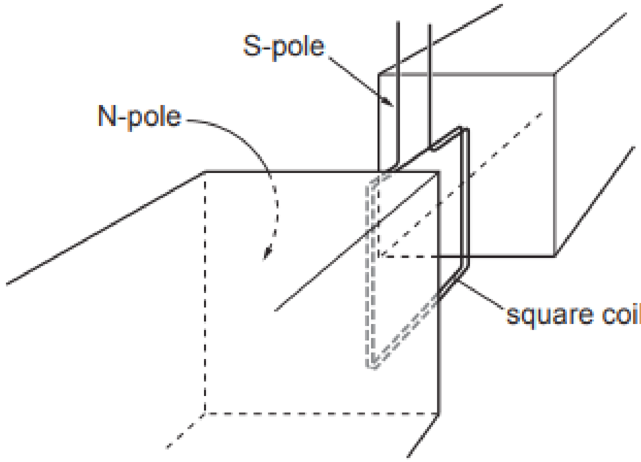


**Fig. 9.4**

Explain why the coil tries to rotate when there is a current in the coil.

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..........................................................................................................................................

.................................................................................................................................... [2]

[Total: 15]

**14. O/N/20/21/Q6**

When electricity is transmitted over large distances, a transformer is used to increase the voltage before transmission. A second transformer is used at the destination to decrease the voltage to the usual mains value.

**(a)** Sketch a labelled diagram to show the structure of a transformer that is used to **increase** voltage.

[2]

**(b)** Describe the principle of operation of a transformer.

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..........................................................................................................................................

.................................................................................................................................... [3]

**(c)** Explain **one** advantage of transmitting electricity at a high voltage.

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..........................................................................................................................................

.................................................................................................................................... [2]

[Total: 7]

**15. M/J/20/22/Q9**

**(a)** Fig. 9.1 shows a simple relay used to switch a mains electric motor on and off.

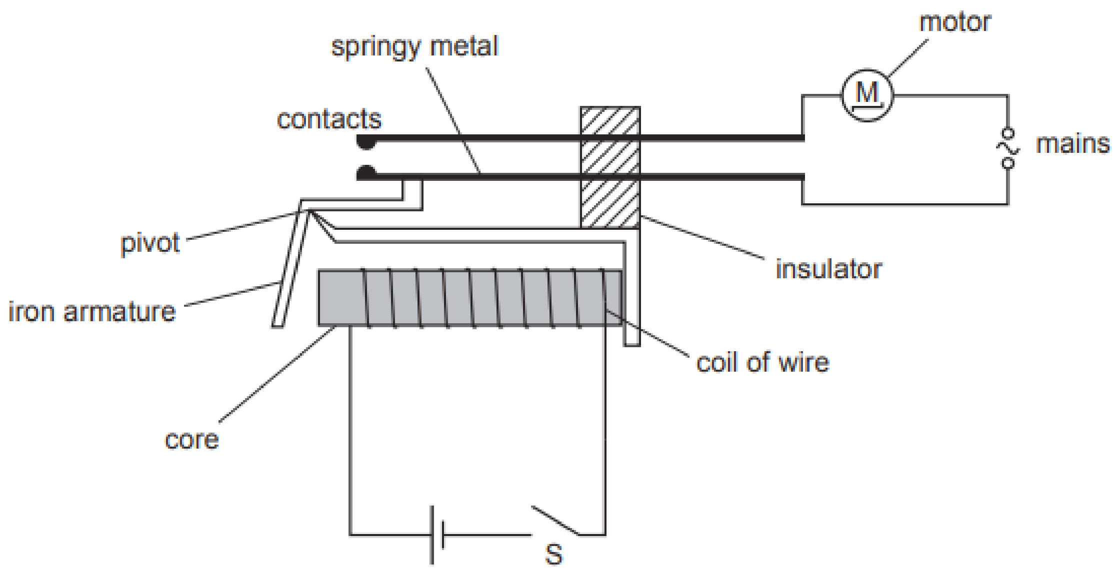


**Fig. 9.1**

**(i)** Explain why the motor switches on when switch S is closed.

..........................................................................................................................

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..........................................................................................................................

.......................................................................................................................... [3]

**(ii)** Explain why the core is made of iron rather than steel.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(iii)** A student suggests that the motor can be turned on and off without a relay.

He suggests connecting the mains to a simple switch in series with the motor.

Suggest **one** reason why, in some situations, using the relay is better.

..........................................................................................................................

.......................................................................................................................... [1]

**(b)** Fig. 9.2 shows the coil of wire wrapped around a cardboard tube. There is no core.

There is an electric current in the wire in the direction shown by the arrows.

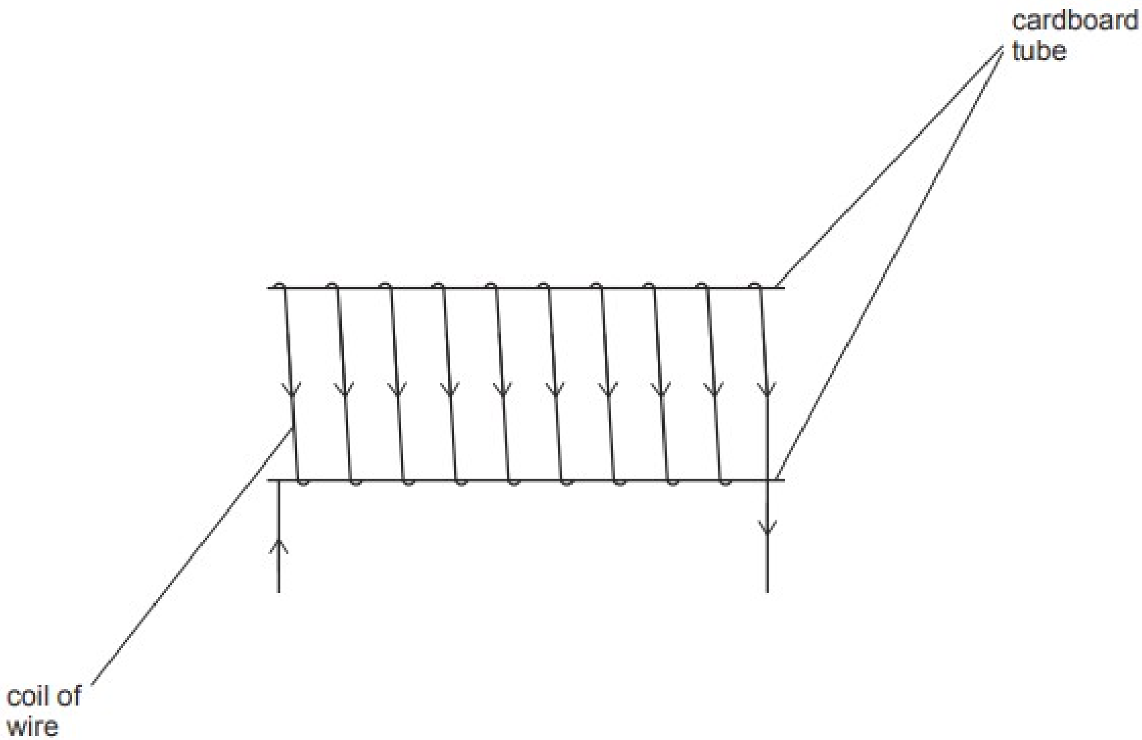


**Fig. 9.2**

**(i)** On Fig. 9.2, draw the pattern of the magnetic field inside and around the coil. Mark the direction of this magnetic field. [4]

**(ii)** On Fig. 9.2, mark the N-pole of the coil. [1]

**(c)** The supply of current to the coil is removed.

The ends of the coil are connected to a sensitive ammeter, as shown in Fig. 9.3.

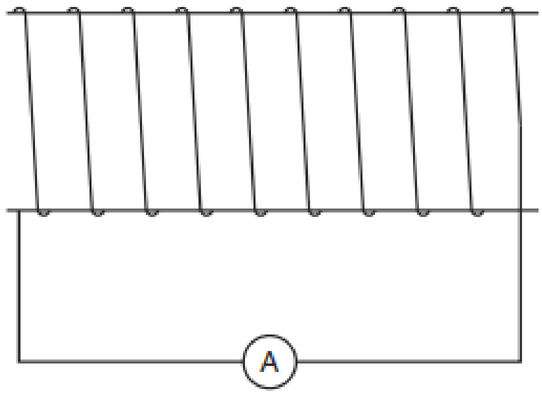


**Fig. 9.3**

**(i)** Describe **how** a permanent magnet is used to produce a **large** reading on the ammeter.

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.................................................................................................................................. [2]

**(ii)** Explain **why** a current is produced in **(i)**.

..........................................................................................................................................

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.................................................................................................................................. [2]

[Total: 15]

**16. M/J/20/21/Q10**

Fig. 10.1 shows a motor lifting a mass. Fig. 10.2 shows part of the circuit diagram of the connections to the motor.

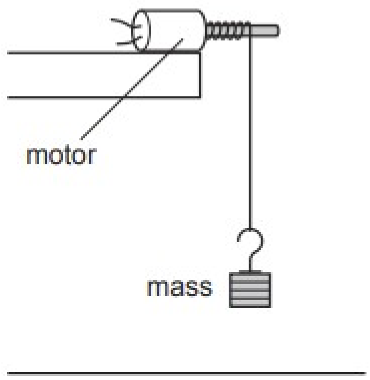


Fig. 10.1

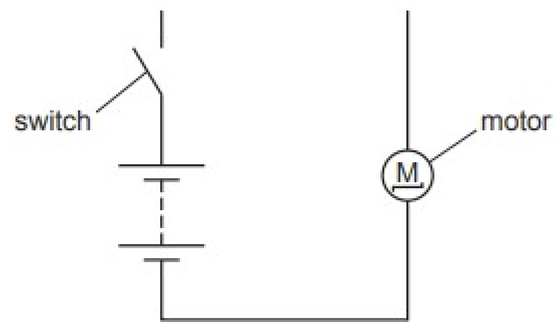


Fig. 10.2

**(a)** The current in the motor is 1.5 A and the voltage supplied by the battery is 8.0 V.

**(i)** Complete the circuit diagram in Fig. 10.2 to show an ammeter and a voltmeter in the correct positions to take these measurements while the motor is working. [2]

**(ii)** The motor takes 4.0 s to lift the mass.

Calculate the electrical energy transferred to the motor in this time.

energy = ............................................................ [2]

**(iii)** The motor lifts the 150 g mass through a height of 80 cm in the 4.0 s.

Calculate the gravitational potential energy gained by the mass.

The gravitational field strength $g$ = 10 N / kg.

gravitational potential energy = ............................................................ [3]

**(iv)** State **two** reasons why the gravitational potential energy gained by the mass is less than the electrical energy supplied to the motor.

1. ............................................................................................................................

2. ............................................................................................................................

[2]

**(b)** Fig. 10.3 shows the structure of the motor.

When the mass reaches the top of its motion, the switch is opened. This disconnects the battery and causes the mass to fall. The coil turns as the mass falls.

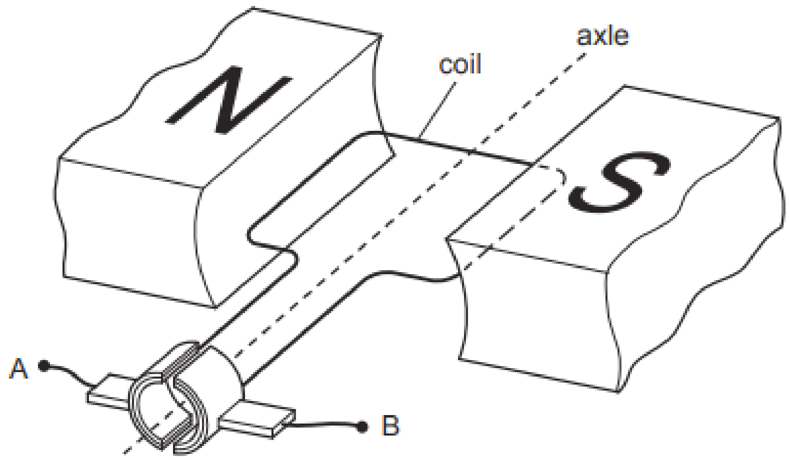


**Fig. 10.3**

As the coil turns, a small voltage is produced.

**(i)** Explain why a voltage is produced as the coil turns.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

**(ii)** As the mass falls, a student connects a wire between the points A and B shown in Fig. 10.3.

He notices that the mass takes a longer time to fall when the wire is connected.

The student suggests that this is an example of Lenz's law.

State Lenz's law and suggest why the mass takes longer to fall.

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.................................................................................................................................... [3]

**17. O/N/19/22//Q6**

An electric circuit contains a 700 Ω resistor and a light-dependent resistor (LDR). Fig. 6.1 is the circuit diagram.

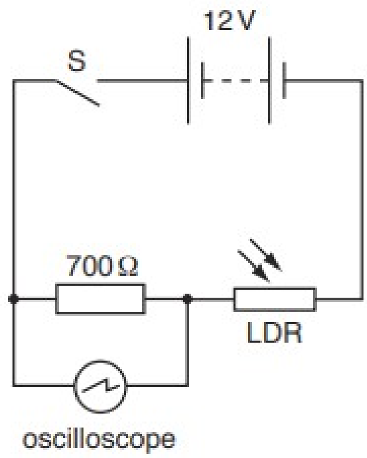


**Fig. 6.1**

The electromotive force (e.m.f.) of the battery is 12 V.

An oscilloscope is connected across the fixed resistor. Fig. 6.2 shows the oscilloscope, including the settings of the timebase and the Y-gain controls. Line Q shows the position of the trace on the oscilloscope when the switch S is open.

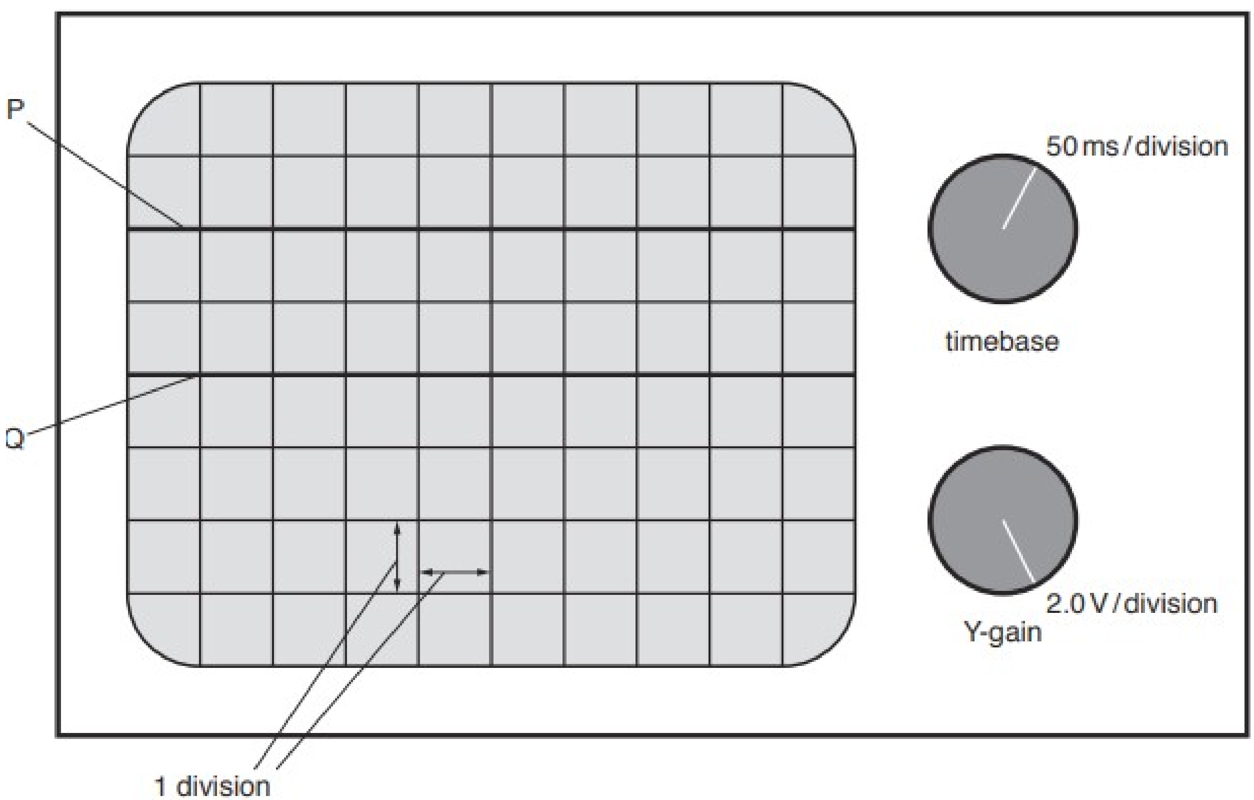


**Fig. 6.2**

The switch S is closed and the trace on the oscilloscope moves to the position shown by line P in Fig. 6.2.

**(a)** **(i)** Determine the potential difference (p.d.) across the 700 Ω resistor.

p.d. = ........................................................ [1]

**(ii)** Determine the resistance of the LDR.

resistance = ........................................................ [3]

**(b)** The intensity of the light incident on the LDR gradually increases.

State and explain how the trace on the oscilloscope screen moves.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

[Total: 7]

## 18. O/N/19/22/Q7

Fig. 7.1 shows a horizontal, rectangular coil ABCD placed between a magnetic N-pole and an S-pole.

There is a current in the coil.

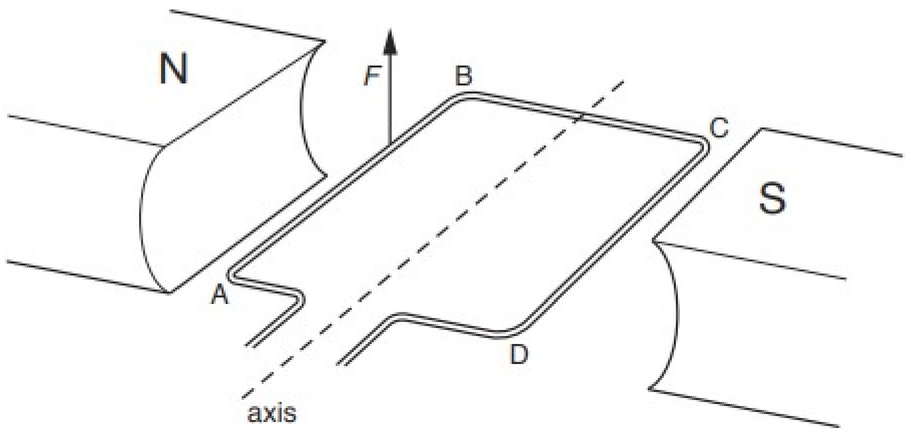


**Fig. 7.1**

An upward force $F$ of size $9.6 \times 10^{-3}$ N acts on side AB of the coil.

**(a)** **(i)** Explain why there is a force on side AB.

.......................................................................................................................................

....................................................................................................................................... [1]

**(ii)** Determine the direction of the current in AB and state how the direction is deduced.

.......................................................................................................................................

.......................................................................................................................................

....................................................................................................................................... [2]

**(b)** Both side AB and side CD of the coil are 2.5 cm from the axis.

Determine the total moment acting on the coil.

moment = ........................................................ [2]

**(c)** The coil in Fig. 7.1 is part of a direct current (d.c.) motor.

**(i)** State the name of the device that connects the coil of a d.c. motor to the electricity supply.

....................................................................................................................................... [1]

**(ii)** State **one** change to the arrangement in Fig. 7.1 that produces a greater turning effect in coil ABCD.

....................................................................................................................................... [1]

## 19. O/N/19/21/Q8

**(a)** A small electric component in a circuit needs to be protected from a strong, external magnetic field.

Describe how this is done and state any material used.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** Electromagnets have many domestic and industrial applications.

**(i)** In the space, draw a diagram of a simple electromagnet.

[3]

**(ii)** State and describe in detail one use of an electromagnet.

..........................................................................................................................................

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..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

(c) Fig. 8.2 shows a solenoid connected to a sensitive ammeter.

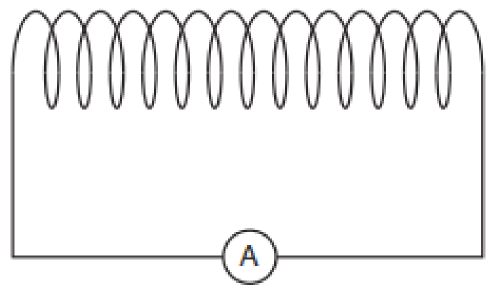


**Fig. 8.2**

Explain what is observed on the ammeter when a student:

(i) inserts an N-pole of a bar magnet slowly into the right-hand end of the solenoid

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................. [3]

(ii) holds the bar magnet stationary in the solenoid

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................. [1]

(iii) withdraws the N-pole of the magnet **quickly** from the right-hand end of the solenoid.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................. [3]

[Total: 15]

**20. M/J/19/22/Q7**

Fig. 7.1 shows a copper wire placed on two copper rods in the magnetic field between the poles of a magnet.

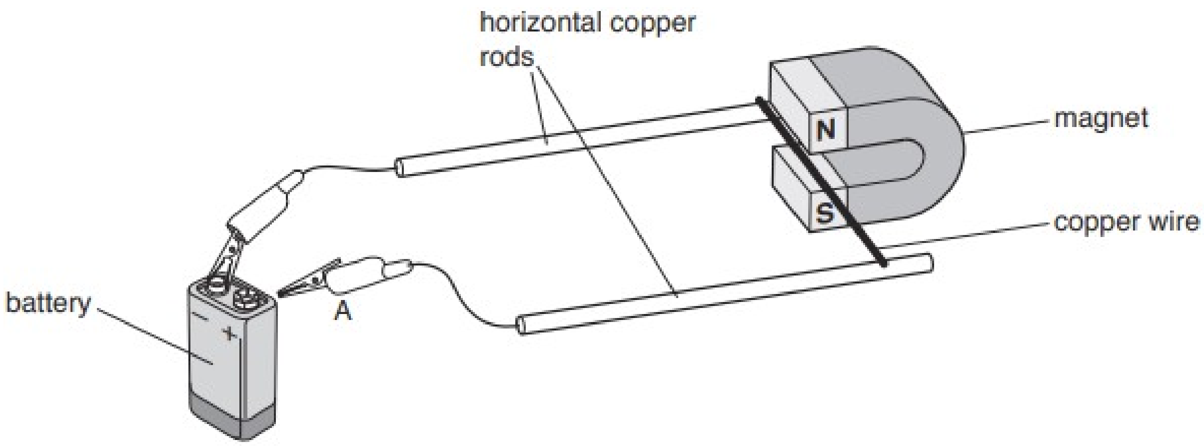


**Fig. 7.1**

The crocodile clip A touches the positive terminal of the battery. This causes the copper wire to move.

**(a)** On Fig. 7.1, mark with an arrow the direction of the current in the copper wire. [1]

**(b)** Explain why the copper wire moves along the copper rods.

.......................................................................................................................................

.......................................................................................................................................

.......................................................................................................................................

....................................................................................................................................... [2]

**(c)** Name **two** different devices that use this effect.

1. .......................................................................................................................................

2. .......................................................................................................................................

[2]

[Total: 5]

**21. M/J/19/21/Q7**

Fig. 7.1 shows part of a torch. The torch does not contain a battery.

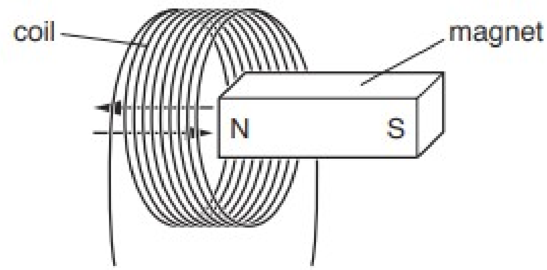


**Fig. 7.1**

The torch is shaken and this causes the magnet to move backwards and forwards through the coil.

**(a)** **(i)** Explain why an electromotive force (e.m.f.) is induced in the coil when the magnet moves.

..........................................................................................................................................

.................................................................................................................................... [2]

**(ii)** State one way to increase the e.m.f. induced.

..........................................................................................................................................

.................................................................................................................................... [1]

**(b)** The current in the coil is used to produce light from a light-emitting diode (LED).

**(i)** In the space below, draw the circuit symbol for an LED.

[1]

**(ii)** An LED is a *more efficient* source of light than a filament lamp.

Explain what this statement means.

..........................................................................................................................................

.................................................................................................................................... [1]

[Total: 5]

**22. O/N/18/22/Q7**

Fig. 7.1 shows the structure of a transformer.

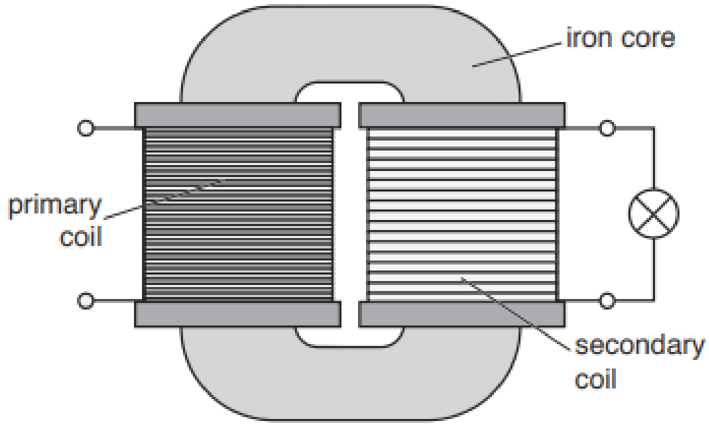


**Fig. 7.1**

A lamp is connected to the secondary coil.

**(a)** Explain why the core of the transformer is made from iron.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**(b)** When there is an alternating current (a.c.) in the primary coil, the lamp is lit.

When there is a direct current (d.c.) in the primary coil, the lamp is **not** lit.

**(i)** State two ways in which an alternating current differs from a direct current.

1. ..........................................................................................................................

..........................................................................................................................

2. ..........................................................................................................................

..........................................................................................................................

[2]

**(ii)** Explain why the lamp is **not** lit when there is a direct current in the primary coil.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

## 23. M/J/18/22/Q7

A computer hard disk contains a layer of a magnetic material.

**(a)** Describe how a magnet is used to find out if a sample of material is magnetic or non-magnetic.

.......................................................................................................................................

.......................................................................................................................................

.......................................................................................................................................[1]

**(b)** Data is stored on the disk as a series of N-poles and S-poles.

Fig. 7.1 shows part of the hard disk. The thin layer of magnetic material contains small regions. Each region has an N-pole and an S-pole. Some magnetic field lines are shown on Fig. 7.1.

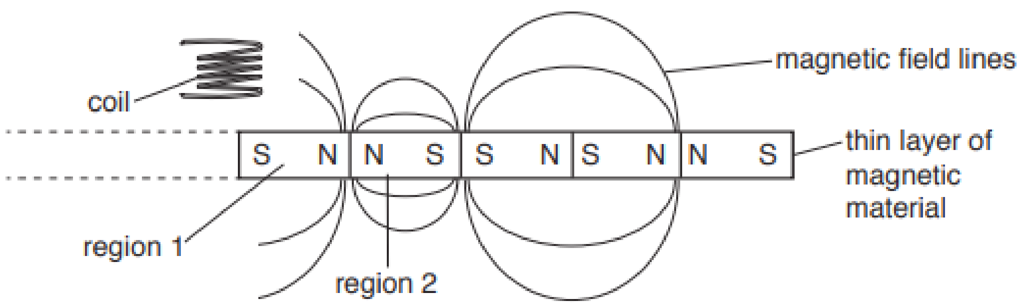


**Fig. 7.1** (not to scale)

**(i)** Region 2 causes a magnetic force on region 1.

State the direction of the magnetic force on region 1 and explain why it acts.

.......................................................................................................................................

.......................................................................................................................................

.......................................................................................................................................[2]

**(ii)** On Fig. 7.1, draw arrows on the field lines to show the direction of the magnetic field near the boundary between region 1 and region 2. [1]

**(iii)** The coil shown in Fig. 7.1 is fixed in position. The layer of magnetic material passes quickly under the coil.

A voltage is induced in the coil as some of the boundaries between the regions pass under the coil.

**1.** Explain why a voltage is induced in the coil.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[1]

**2.** Suggest why the coil must be close to the layer.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[1]

**24. O/N/17/22/Q11**

The metals in the list below have many different uses.

**aluminium** **copper** **iron** **silver** **steel**

**(a)** State which metal from the list is used for

**(i)** a compass needle,

..........................................................................................................................................

**(ii)** magnetic screening,

..........................................................................................................................................

**(iii)** the core of a transformer.

..........................................................................................................................................
[2]

**(b)** **(i)** Describe one use for a transformer.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................[2]

**(ii)** An a.c. generator supplies an input voltage of 220 V to a transformer.

**1.** In the space below, sketch a graph of the output voltage against time for the a.c. generator. [2]

**2.** The transformer has 1700 turns on the primary coil and 85 turns on the secondary coil.

Calculate the output voltage of the transformer.

output voltage = ..........................................................[2]

**(c)** Fig. 11.1 shows a magnet next to one end of a solenoid.

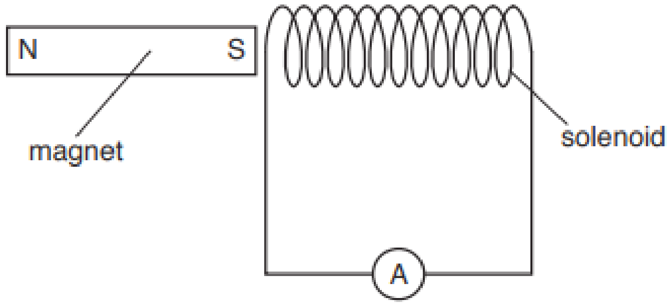


**Fig. 11.1**

The terminals of the solenoid are connected to a very sensitive ammeter.

**(i)** The magnet is moved to the right at a constant speed and a reading is observed on the ammeter.

**1.** Explain why there is a current in the ammeter.

..........................................................................................................................

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..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[3]

**2.** Explain how the current in the ammeter opposes the change producing it.

..........................................................................................................................

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..........................................................................................................................

..........................................................................................................................[2]

**(ii)** The magnet stops when the S-pole reaches the middle of the solenoid.

The reading on the ammeter is observed when the magnet is moved to the left at a constant speed that is less than its speed in **(c)(i)**.

State two ways in which the reading on the ammeter differs from the reading observed in **(c)(i)** as the magnet moves to the left.

1. ..........................................................................................................................

2. ..........................................................................................................................

[2]

## 25. O/N/17/21/Q7

An electric circuit consists of a battery, a variable resistor and connecting wires. Sections PQ and QR of the connecting wire are in a region of space where there is a uniform magnetic field.

In Fig. 7.1, the shaded area represents the magnetic field.

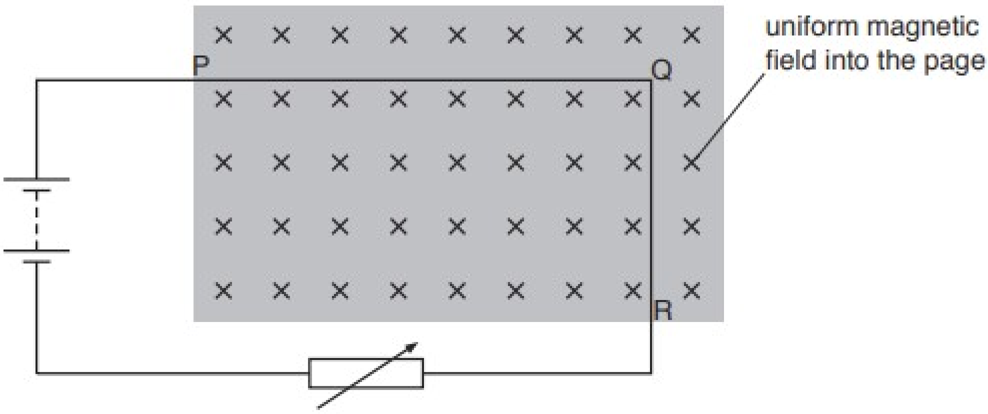


**Fig. 7.1**

The direction of the magnetic field is into the page.

A current-carrying wire in a magnetic field can experience a force.

**(a)** In the table, mark **one** tick (✓) in each column to indicate the force on PQ and the force on QR.

| | section PQ | section QR |
|---|---|---|
| no force on this section | | |
| a force towards the top of the page | | |
| a force towards the bottom of the page | | |
| a force out of the page | | |
| a force into the page | | |
| a force towards the left of the page | | |
| a force towards the right of the page | | |

[2]

**(b)** The battery in the circuit in Fig. 7.1 is reversed.

State and explain the effect of this on PQ and QR.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**(c)** The variable resistor is adjusted and the current in the circuit decreases.

Suggest one effect of decreasing the current.

..........................................................................................................................................

.................................................................................................................................... [1]

**26. M/J/17/22/Q7**

Fig. 7.1 shows part of a d.c. electric motor.

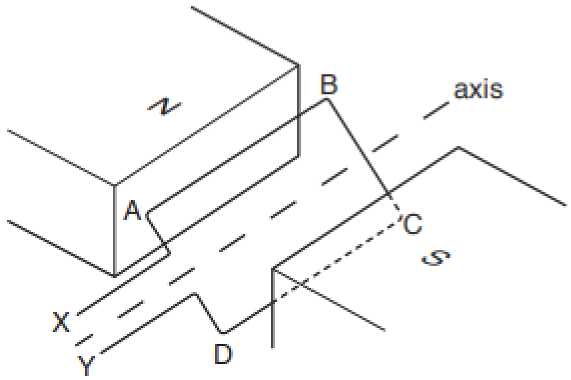


**Fig. 7.1**

**(a)** A coil of wire ABCD is placed between the poles of a magnet. Ends X and Y are connected to a commutator and a battery.

**(i)** State why there is a force on side AB of the coil.

..........................................................................................................................................

.......................................................................................................................................[1]

**(ii)** Describe how the commutator keeps the coil rotating in one direction.

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..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................[2]

**(b)** The current in the motor is 2.0 A and the battery has an electromotive force (e.m.f.) of 12 V. In a time of 8.0 s, the motor does 140 J of work.

Calculate

**(i)** the energy supplied to the motor in 8.0 s,

energy supplied = ..............................................................[2]

**(ii)** the efficiency of the motor.

efficiency = ..............................................................[1]

**27. M/J/17/21/Q7**

Fig. 7.1 shows a metal rod placed between the poles of a magnet.

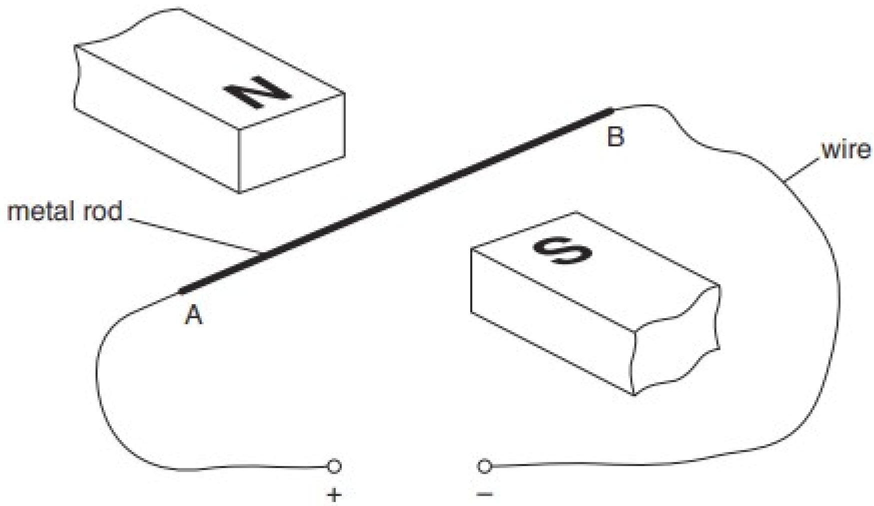


**Fig. 7.1**

**(a)** End A of the rod is connected to the positive terminal of a d.c. power supply and end B is connected to the negative terminal. The current in the rod and the magnetic field produce a force on the rod.

On Fig. 7.1 draw arrows to show

**(i)** the direction of the current in the metal rod; label this arrow C,

**(ii)** the direction of the magnetic field that acts on the metal rod; label this arrow M,

**(iii)** the direction of the force on the rod; label this arrow F.

[3]

**(b)** The power supply is removed and a sensitive voltmeter is connected between the ends A and B.

When the rod is moved upwards or downwards there is a reading on the voltmeter.

**(i)** Explain why there is a reading on the voltmeter.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................[2]

**(ii)** The reading on the voltmeter is increased by using a stronger magnetic field.

State one other way of producing a larger reading on the voltmeter.

..........................................................................................................................................

..........................................................................................................................................[1]

**28. O/N/16/21/Q6**

Two flexible iron strips, WX and YZ, are placed close to each other inside a solenoid (long coil). The end W of WX and the end Z of YZ are held firmly in position.

Fig. 6.1 shows that the solenoid is connected to a d.c. power supply and a switch.

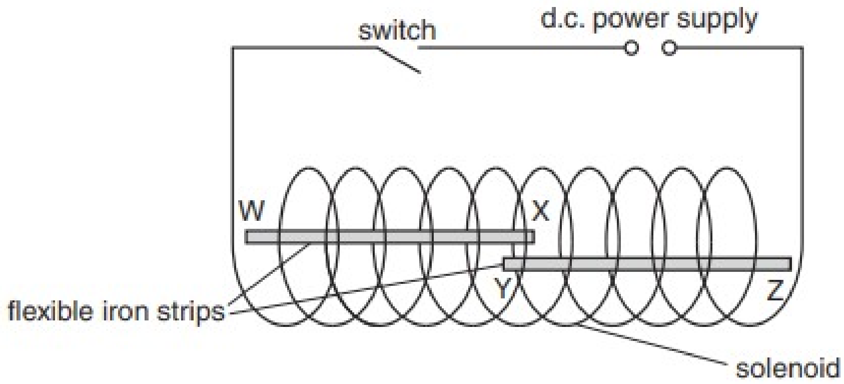


**Fig. 6.1**

The switch is closed and there is an electric current in the solenoid.

End W of flexible iron strip WX becomes a magnetic S-pole.

**(a)** **(i)** State the type of magnetic pole produced at X, Y and Z.

**1.** X ........................-pole

**2.** Y ........................-pole

**3.** Z ........................-pole

[2]

**(ii)** State and explain what happens to X and Y because the flexible iron strips are magnetised.

..........................................................................................................................

..........................................................................................................................[1]

**(b)** A relay operates a switch in one circuit using the current in a different circuit.

State one example of a relay used in this way and explain why a relay is used.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[2]

**29. M/J/16/22/Q8**

A transformer and a diode are used to charge a battery.

Fig. 8.1 shows the transformer, which contains a soft-iron core and two coils.

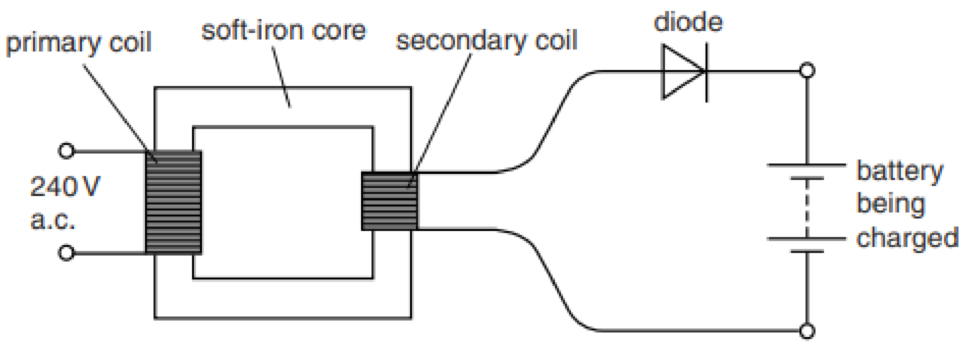


**Fig. 8.1**

The primary coil is connected to the 240 V a.c. mains supply. The secondary coil is connected in series with the diode and the battery.

**(a)** Explain why an electromotive force (e.m.f.) is induced in the secondary coil.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................[2]

**(b)** The e.m.f. induced in the secondary coil is less than 240 V.

Suggest why.

..........................................................................................................................................

.......................................................................................................................................[1]

**(c)** Suggest why steel is not used as the core of a transformer.

..........................................................................................................................................

.......................................................................................................................................[1]

**(d)** Describe the action of the diode.

..........................................................................................................................................

.......................................................................................................................................[1]

**30. M/J/16/21/Q7**

Fig. 7.1 shows part of a loudspeaker. A coil is attached to a paper cone and is suspended freely within a cylindrical magnet.

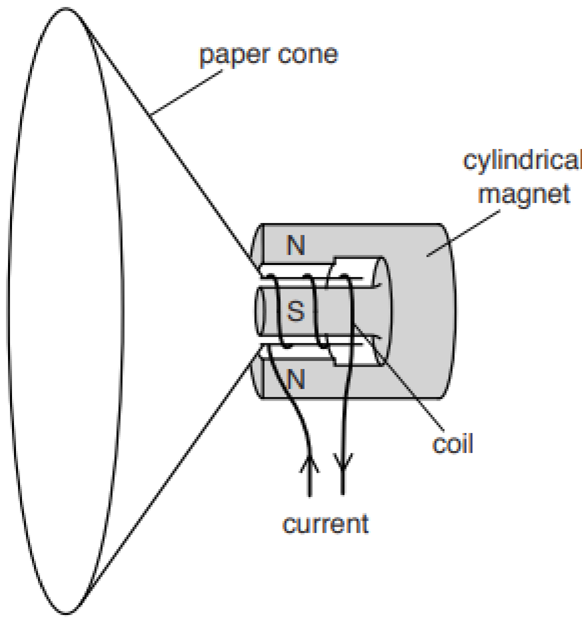


**Fig. 7.1**

**(a)** Explain why a current in the coil causes the paper cone to move.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(b)** Describe and explain the movement of the cone when there is an alternating current in the coil.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(c)** The loudspeaker produces sound of frequency 500 Hz. The speed of sound is 320 m/s. Calculate the wavelength of the sound produced.

wavelength = ..........................................................[2]

**31. M/J/16/21/Q11**

**(a)** Induced currents can be formed in a coil of wire by electromagnetic induction.

**(i)** Describe how to demonstrate the formation of an induced current in a coil of wire.

Sketch and label a diagram of the arrangement of the apparatus.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[3]

**(ii)** State two changes to the apparatus that increase the induced current.

1. ..........................................................................................................................

2. ..........................................................................................................................
[2]

**(iii)** The direction of the induced current is determined by Lenz's Law.

**1.** State Lenz's Law.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[2]

**2.** Describe how Lenz's Law applies in the experiment you described in **(a)(i)**.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[2]

**(b)** An electrical generator supplies power to a distant factory.

**(i)** The factory receives 500 kW of electrical power at a voltage of 33 kV.

Calculate

**1.** the current in the wires supplying the power,

current = ............................................................[2]

**2.** the electrical energy supplied to the factory in one hour.

energy = ............................................................[2]

**(ii)** The 33 kV voltage supplied to the factory is very high.

Transmitting electrical energy at high voltage reduces energy losses in the transmission lines.

Explain why.

.......................................................................................................................................

.......................................................................................................................................

.......................................................................................................................................

.......................................................................................................................................

.......................................................................................................................................[2]

## 32. O/N/15/22/Q10

Thin wire, covered in plastic insulation, is used to make a solenoid (long coil). The solenoid is connected to a sensitive ammeter. Fig. 10.1 shows the N-pole of a steel magnet placed next to the solenoid.

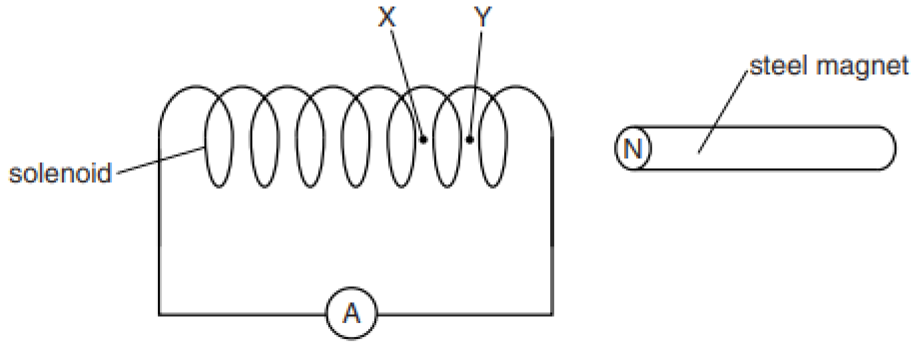


**Fig. 10.1**

Point X and point Y are on the axis of the solenoid.

**(a)** **(i)** Explain why plastic is an electrical insulator.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [1]

**(ii)** Explain why the magnet is **not** made from

**1.** aluminium,

.......................................................................................................................... [1]

**2.** iron.

.......................................................................................................................... [1]

**(b)** In one experiment, the magnet in Fig. 10.1 is moved to the left and passes into the solenoid. The N-pole of the magnet travels from Y to X at a constant speed. As it moves, the ammeter shows a small current.

**(i)** Explain why there is a current in the solenoid when the magnet is moving.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [3]

**(ii)** The N-pole travels from Y to X in 0.14 s. As it moves, the current shown on the ammeter is 0.045 mA. The resistance of the solenoid is 1.2 Ω.

Calculate

**1.** the potential difference (p.d.) across the solenoid,

potential difference = ........................................................ [2]

**2.** the charge that passes through the solenoid as the N-pole moves from Y to X.

charge = ........................................................ [2]

**(c)** In a second experiment, the speed of the N-pole is greater than its speed in the first experiment. It now takes only 0.070 s to travel from Y to X.

A current in the same direction is shown on the ammeter.

**(i)** State and explain how the size of this current compares with the size of the current in the first experiment.

...........................................................................................................................

...........................................................................................................................

........................................................................................................................... [2]

**(ii)** The same quantity of charge passes through the coil in both the first and second experiments.

Explain why this is the case.

...........................................................................................................................

...........................................................................................................................

........................................................................................................................... [1]

**(d)** State two ways in which the equipment shown in Fig. 10.1 can be used to produce a current in the solenoid that is in the opposite direction.

1. ...........................................................................................................................

2. ...........................................................................................................................

[2]

**33. O/N/15/21/Q10**

**(a)** Fig. 10.1 shows a solenoid (long coil) X connected in series with a battery, a switch S and a variable resistor (rheostat).

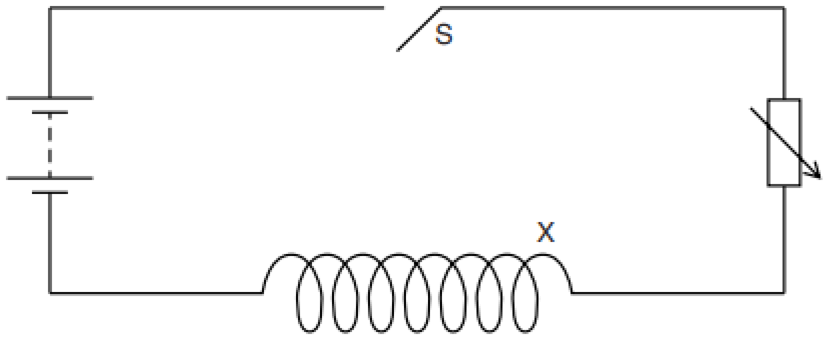


**Fig. 10.1**

The switch S is closed and there is a magnetic field due to the current in the solenoid.

**(i)** On Fig. 10.1, draw the pattern of the magnetic field in, above and below the solenoid. [3]

**(ii)** A second solenoid Y is connected to a sensitive centre-zero ammeter. Solenoid Y is placed in the magnetic field of X.

**1.** The resistance of the variable resistor is gradually decreased and the pointer of the sensitive ammeter deflects slightly to one side.

Explain why this happens.

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2. The switch S is now opened.

Describe and explain what happens to the deflection on the ammeter as the switch is opened.

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**(b)** Fig. 10.2 shows a transformer that consists of two coils wound on an iron core. The transformer is connected to an electricity transmission cable.

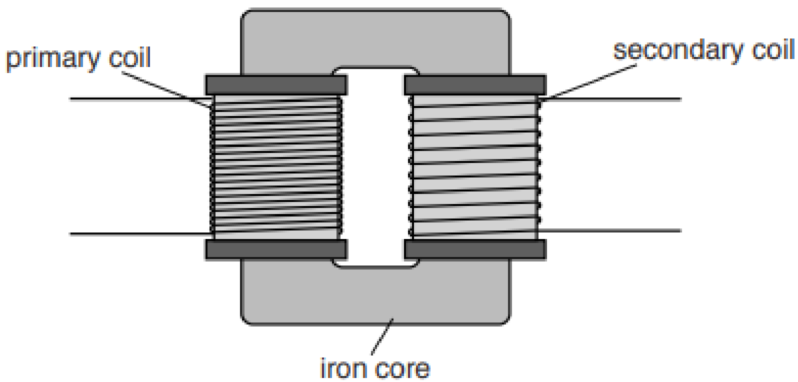


**Fig. 10.2**

**(i)** Explain the purpose of the iron core in the transformer.

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............................................................................................................................ [2]

**(ii)** The transformer supplies electrical power to a factory at 33 000 V. The current supplied is 85 A.

Calculate

**1**. the electrical power that the factory is receiving,

power = ........................................................ [2]

**2**. the electrical energy the factory receives in 1.0 hour.

energy = ........................................................ [2]

**34. M/J/15/22/Q6**

Fig. 6.1 shows a simple a.c. generator.

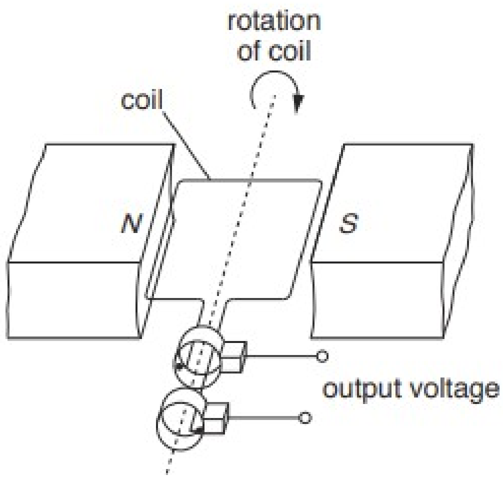


**Fig. 6.1**

**(a)** The coil rotates and an alternating electromotive force (e.m.f.) is induced in the coil.

Fig. 6.2 shows how the alternating e.m.f. varies with time as the coil rotates.

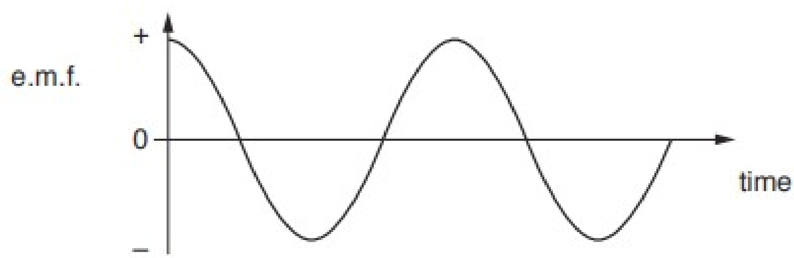


**Fig. 6.2**

Explain

**(i)** why an e.m.f. is induced,

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..........................................................................................................................

.................................................................................................................... [2]

**(ii)** why the e.m.f. is sometimes positive and sometimes negative.

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.................................................................................................................... [1]

**(b)** Changes are made to the a.c. generator, one at a time:

- stronger magnets are used
- more turns are wound on the coil
- the coil is turned faster.

Complete the table in Fig. 6.3 to show what happens to the maximum value of the e.m.f. and to the frequency of the alternating e.m.f.

| changes made | what happens to the maximum value of the e.m.f. | what happens to the frequency of the e.m.f. |
|---|---|---|
| stronger magnets | | |
| more turns on the coil | | |
| the coil is turned faster | | |

**Fig. 6.3**

[3]

## 35. M/J/15/22/Q10

Fig. 10.1 shows a relay connected to a cell and a switch.

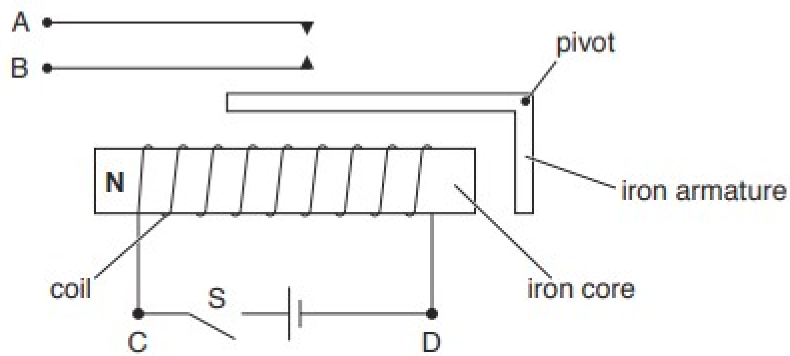


**Fig. 10.1**

**(a)** When switch S is closed, the iron core is magnetised. The left side of the core is an N-pole, as shown in Fig. 10.1. The iron armature is attracted to the core.

**(i)** On Fig. 10.1, mark

**1.** the S-pole of the iron core

**2.** the N-pole and the S-pole of the iron armature.

[2]

**(ii)** The cell is reversed.

State what happens to the poles and to the armature.

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.......................................................................................................................... [2]

**(iii)** Explain why the core is made of iron and not steel.

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..........................................................................................................................

.......................................................................................................................... [2]

**(b)** Fig. 10.2 shows the relay connected in a circuit to a 12 V battery. The bell is not ringing.

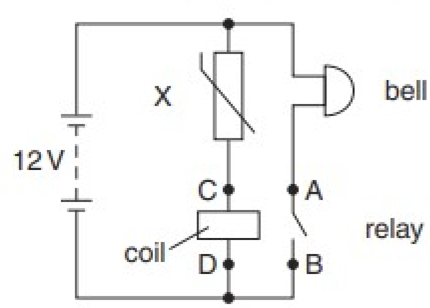


**Fig. 10.2**

**(i)** State the name of component X.

.......................................................................................................................... [1]

**(ii)** Explain why the bell rings when the temperature of X rises.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(iii)** When the resistance of X is $2000\,\Omega$, the current in the coil is 1.5 mA. This causes the contacts in the relay to close. The resistance of the bell is $200\,\Omega$.

Calculate

**1.** the potential difference (p.d.) across X,

p.d. = ............................................................ [2]

**2.** the p.d. across the coil,

p.d. = ............................................................ [1]

**3.** the current in the battery.

current = ............................................................ [2]

**(iv)** Component X is removed from the circuit and replaced by a different component Y. The bell now rings when bright light shines on Y.

State the name of component Y.

.......................................................................................................................... [1]

**36. M/J/15/21/Q7**

A simple apparatus used to demonstrate electromagnetic induction is shown in Fig. 7.1.

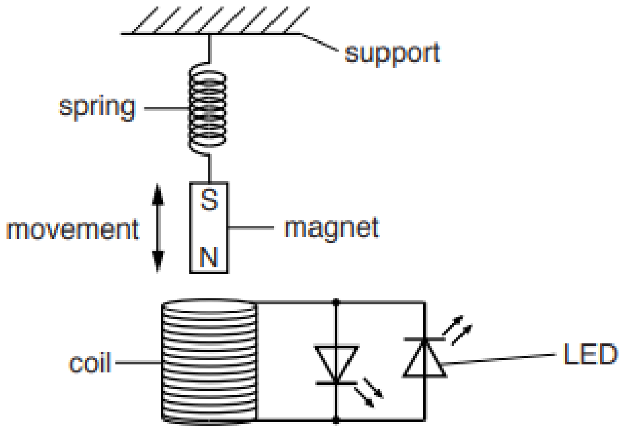


**Fig. 7.1**

The coil is connected to two light-emitting diodes (LEDs). The magnet moves into and out of the coil.

**(a)** Explain why there is an induced e.m.f. in the coil when the magnet moves.

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**(b)** Explain why one LED lights up when the magnet moves into the coil and the other LED lights up when the magnet moves out of the coil.

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..........................................................................................................................................[2]

**(c)** The LEDs are brighter when the magnet moves faster.
Explain why.

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## 37. O/N/14/22/Q8

Fig. 8.1 shows the structure of a simple alternating current (a.c.) generator.

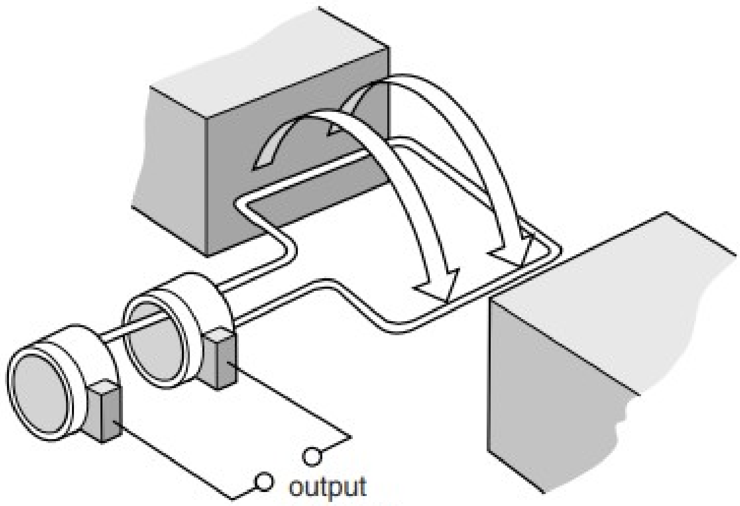


**Fig. 8.1**

**(a)** On Fig. 8.1, label

- the coil of the generator with the letter C,
- a slip ring with the letter S,
- a carbon brush with the letter B.

[2]

**(b)** The a.c. generator is operating and the arrows on Fig. 8.1 show the direction of rotation.

Explain why there is an electromotive force (e.m.f.) between the two output terminals.

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..........................................................................................................................................[3]

**(c)** The output terminals of the a.c. generator are connected to a cathode-ray oscilloscope (c.r.o.). Fig. 8.2 shows the trace on the screen of the c.r.o.

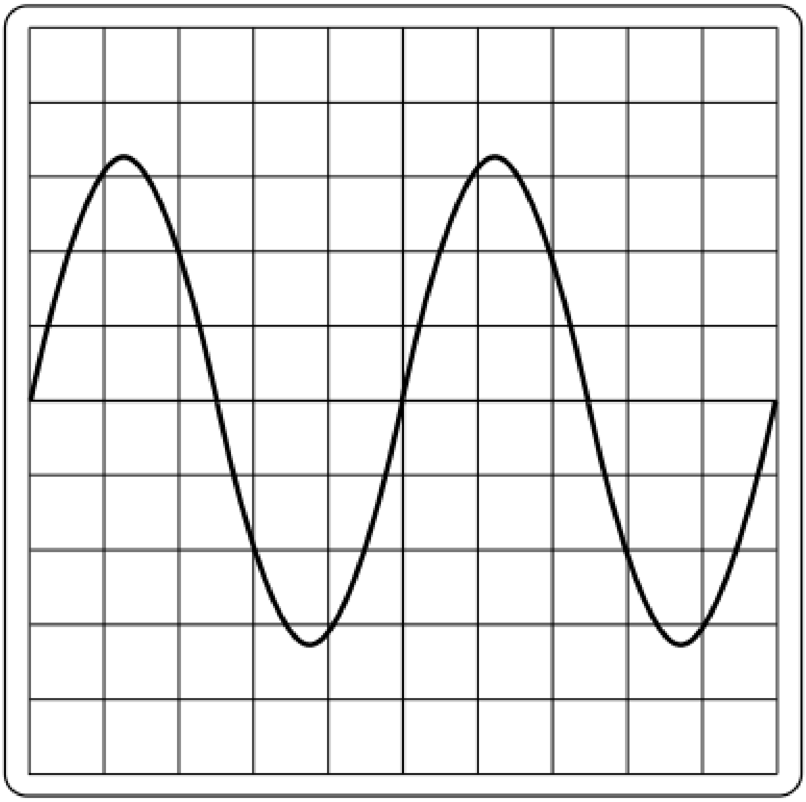

**Fig. 8.2**

Describe how the trace and a setting on the c.r.o. are used to find the time for one revolution of the coil of the a.c. generator. You may draw on Fig. 8.2 if you wish.

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..........................................................................................................................................

......................................................................................................................................[2]

**38. O/N/14/21/Q7**

A straight length of copper wire lies horizontally between the poles of a U-shaped magnet. Fig. 7.1 shows the two ends of the wire connected to a very sensitive, centre-zero ammeter.

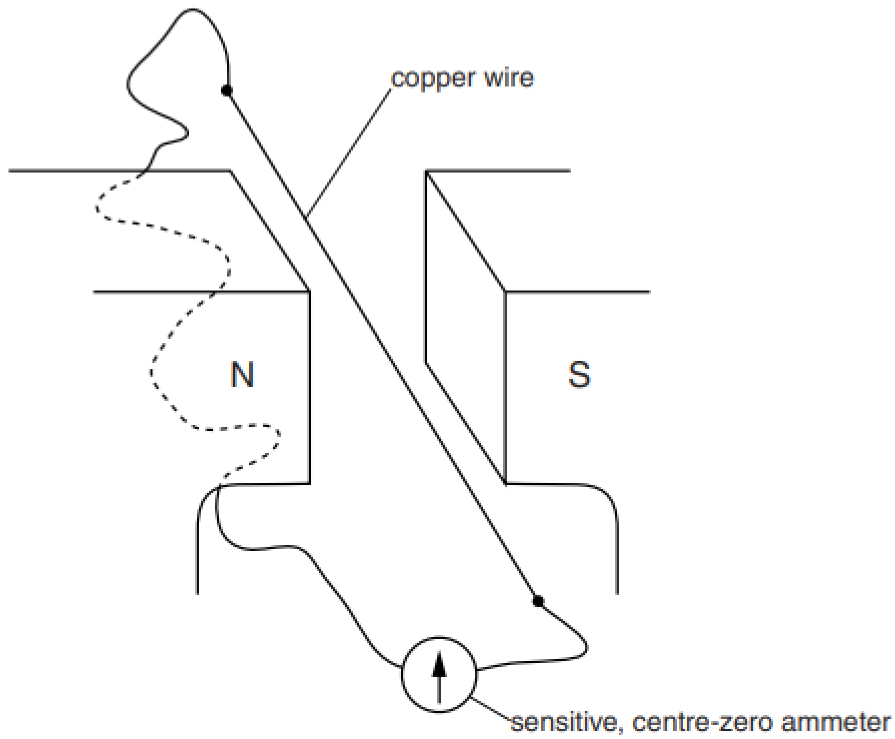


**Fig. 7.1**

The copper wire is moved upwards slowly between the two magnetic poles. The needle on the ammeter deflects to the right.

**(a)** Explain why the needle on the ammeter deflects.

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..........................................................................................................................................[2]

**(b)** The wire is moved downwards very quickly between the two magnetic poles.

State what happens to the needle on the ammeter.

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**(c)** State what happens to the needle on the ammeter when the copper wire is moved horizontally between the two poles.

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**39. M/J/14/21/Q6**

Fig. 6.1 shows a coil of wire connected by flexible leads to a switch and a battery.

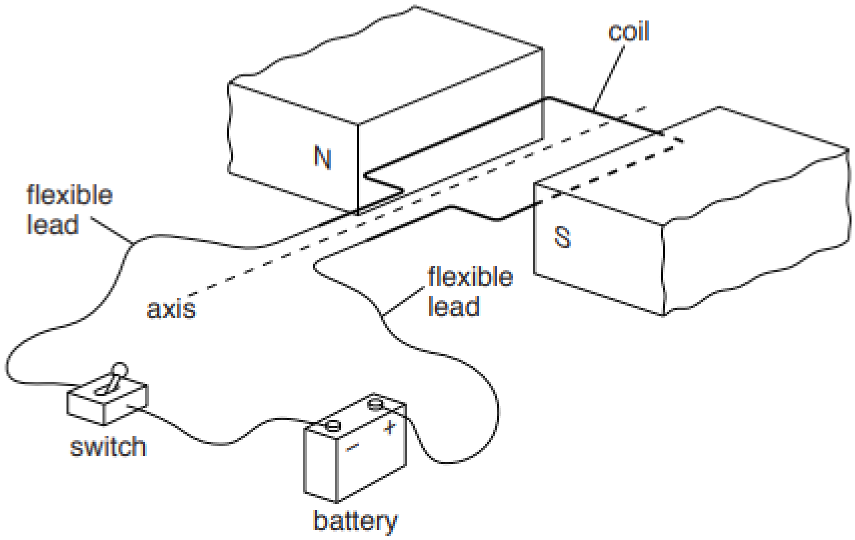


**Fig. 6.1**

The coil is placed between the poles of a permanent magnet and is free to turn about the axis.

When the switch is closed, forces due to the current act on the sides of the coil. The coil starts to turn.

**(a)** On Fig. 6.1, draw arrows to show the directions of the forces. [2]

**(b)** The coil stops when it is vertical. Explain why the turning effect of the forces is zero at this position.

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.................................................................................................................................... [1]

**(c)** In order for the coil to rotate continuously, a split-ring commutator is connected between the battery and the coil.

Explain how the split-ring commutator enables the coil to rotate continuously. Include a diagram in your answer.

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**40. M/J/13/22/Q7**

Fig. 7.1 shows a view, from above, of two wires X and Y. These wires carry equal currents vertically downwards through a piece of card.

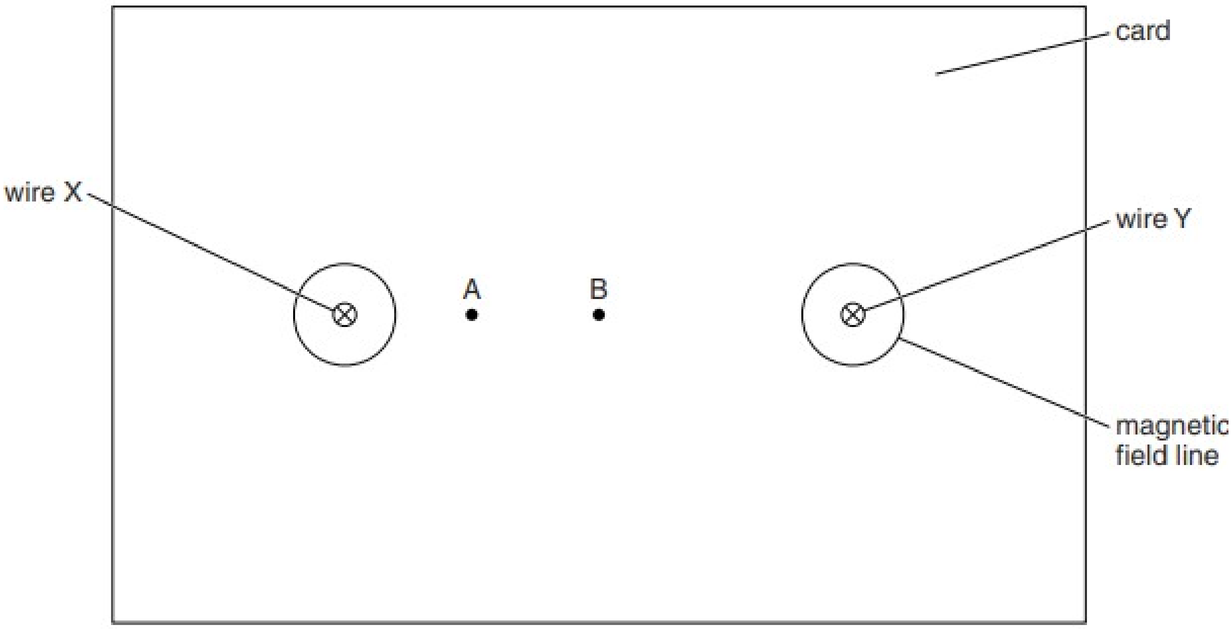


**Fig. 7.1**

One complete magnetic field line is drawn around each wire.

In this question, ignore the effects of the Earth's magnetic field.

**(a)** On Fig. 7.1, draw the complete magnetic field line due to the current in wire X that passes through point A.
Mark the direction of this field line. [2]

**(b)** Point B is midway between the two wires. Explain why the magnetic field at B is zero.

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.......................................................................................................................................... [1]

**(c)** There is a force on wire Y due to the current in wire X.

**(i)** State the direction of the force on wire Y.

.................................................................................................................................. [1]

**(ii)** Explain why there is a force on wire Y.

..................................................................................................................................

.................................................................................................................................. [1]

**41. M/J/13/22/Q8**

Fig. 8.1 shows a simple a.c. generator. The coil is turning and an e.m.f. is induced in the coil.

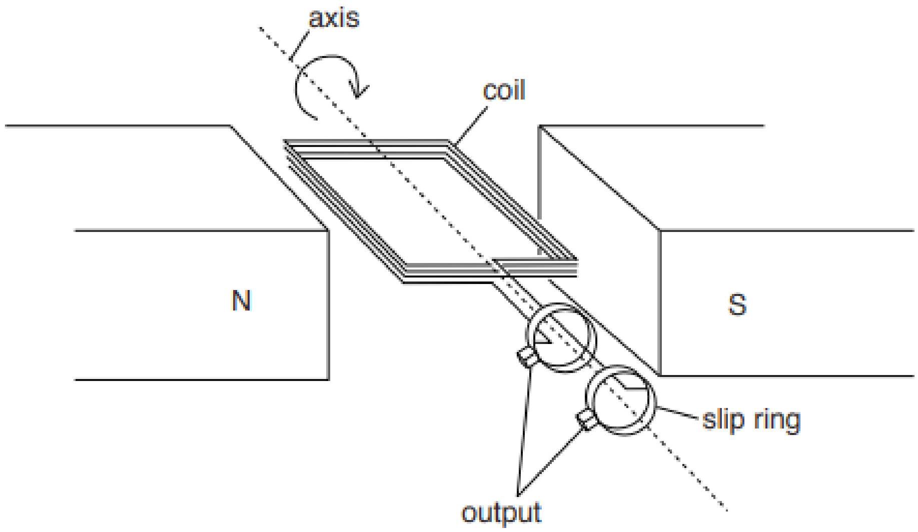


**Fig. 8.1**

**(a)** The generator contains a permanent magnet. State the name of a metal used in a permanent magnet.

.......................................................................................................................... [1]

**(b)** At the instant shown in Fig. 8.1, the induced e.m.f. is a maximum.

**(i)** Explain why the induced e.m.f. is a maximum.

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.......................................................................................................................... [2]

**(ii)** State the position of the coil where there is no induced e.m.f.

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.......................................................................................................................... [1]

**42. M/J/13/21/Q8**

Fig. 8.1 shows a simple transformer.

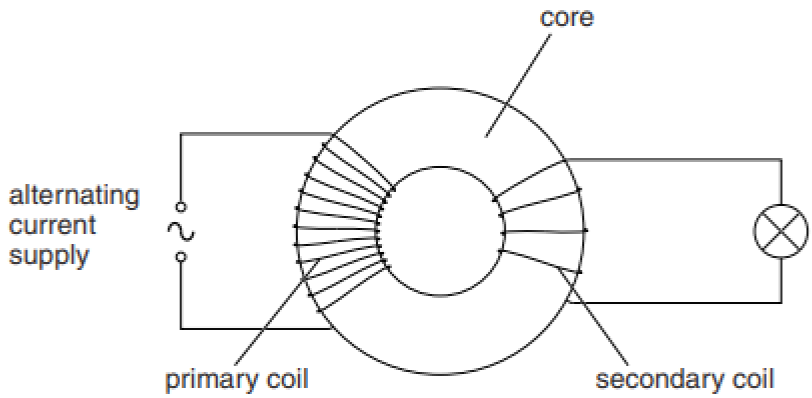


**Fig. 8.1**

**(a)** State the metal used for the core of a transformer.

.......................................................................................................................... [1]

**(b)** Explain how an alternating current in the primary coil causes the lamp to light.

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.......................................................................................................................... [3]

**(c)** Transformers are used to produce high voltages for the transmission of electrical power over long distances.

State one advantage of high voltage transmission.

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**43. O/N/12/22/Q11**

**(a)** A wire is wound around a soft-iron core forming a solenoid, as shown in Fig. 11.1.

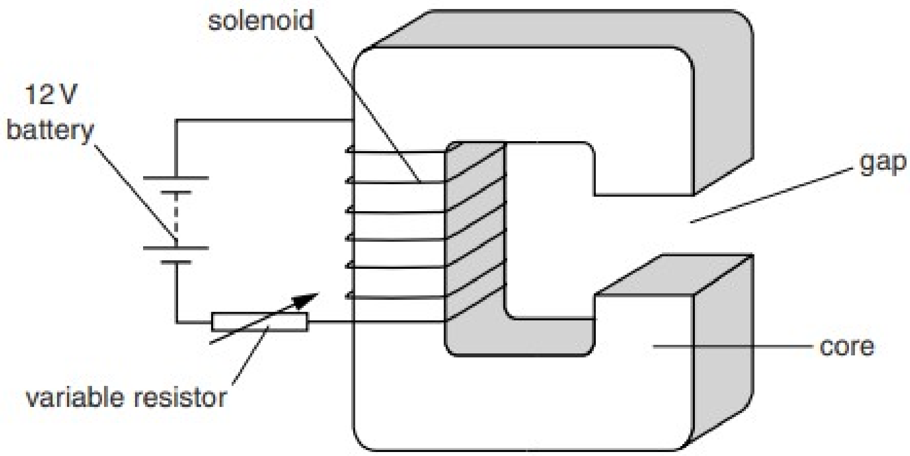


**Fig. 11.1**

There is a gap in the core. The solenoid is connected in series with a 12 V battery and a variable resistor (rheostat). The resistance of the solenoid is 0.30 Ω and the variable resistor is set so that it has a resistance of 4.5 Ω.

**(i)** Calculate the current in the solenoid.

current = ........................................................ [3]

**(ii)** The current in the solenoid magnetises the soft-iron core.

Explain how the electric circuit is used to increase the strength of the magnetic field.

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.................................................................................................................... [2]

**(iii)** Fig. 11.2 shows a horizontal, current-carrying wire PQ in the gap.

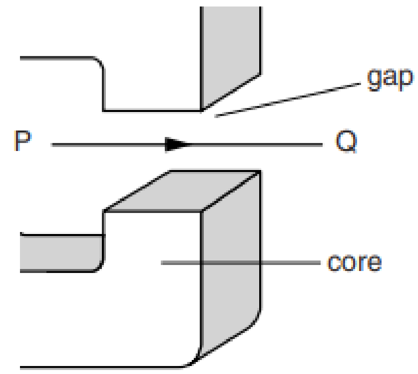


**Fig. 11.2**

**1.** The magnetic field in the gap is uniform and vertically upwards. The current in PQ is from left to right. Describe the effect of the magnetic field on PQ.

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.......................................................................................................................[2]

**2.** State the effect on PQ of increasing the strength of the magnetic field in the gap.

..........................................................................................................................

.......................................................................................................................[1]

**(b)** The starter motor in a car is powered by a 12V battery that is positioned next to the motor. The current in the motor is 75 A.

**(i)** Calculate the power supplied by the battery.

power = ..................................................[2]

**(ii)** Suggest and explain why the wires that connect the motor to the battery are very thick.

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.......................................................................................................................[2]

**(c)** A relay is used to switch on a starter motor in a car. Describe how the relay works. You may include a diagram in your answer.

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.......................................................................................................................................[3]

## 44. O/N/12/21/Q7

**(a)** Fig. 7.1 shows a solenoid made from wire wound around a plastic cylinder.

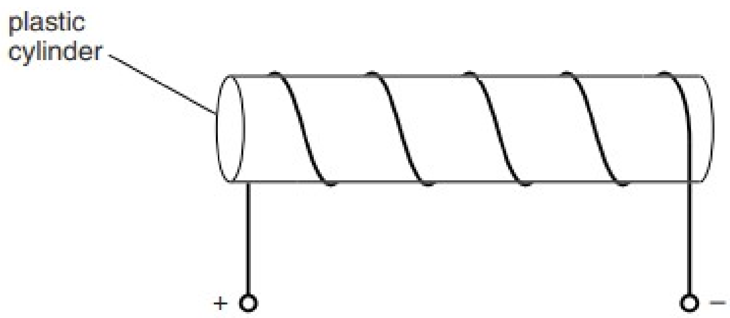


**Fig. 7.1**

A current in the solenoid produces a magnetic field.

On Fig. 7.1, draw the pattern of the magnetic field lines inside and outside the cylinder. [2]

**(b)** Fig. 7.2 shows a beam of beta-particles, in a vacuum, passing into a magnetic field.

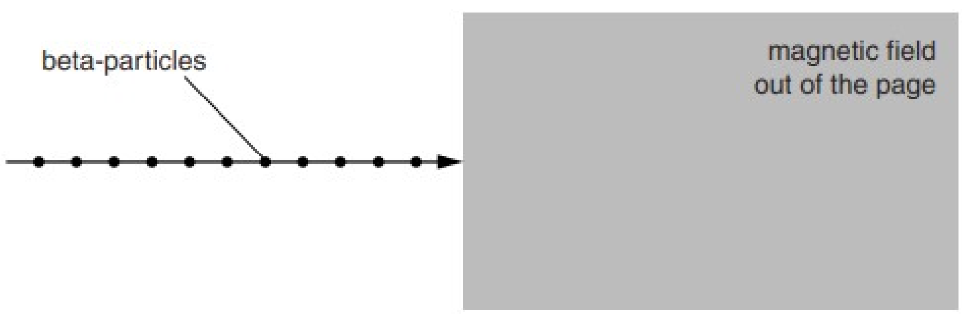


**Fig. 7.2**

The movement of the beta-particles from left to right is an electric current.

**(i)** On Fig. 7.2, draw an arrow to show the direction of the conventional current. [1]

**(ii)** A solenoid is used to produce the magnetic field that lies within the shaded region of Fig. 7.2. The direction of the field is out of the page.

**1.** On Fig. 7.2, draw the path followed by one of the beta-particles in the shaded region. [2]

**2.** The direction of the current in the solenoid is reversed. State what happens to the path of the beta-particle.

..........................................................................................................................

.......................................................................................................................... [1]

## 45. M/J/12/22/Q9

Fig. 9.1 shows two coils of insulated wire wound on an iron ring. Coil A is connected to a battery and a switch. The switch is open. Coil B is connected to a sensitive centre-zero voltmeter.

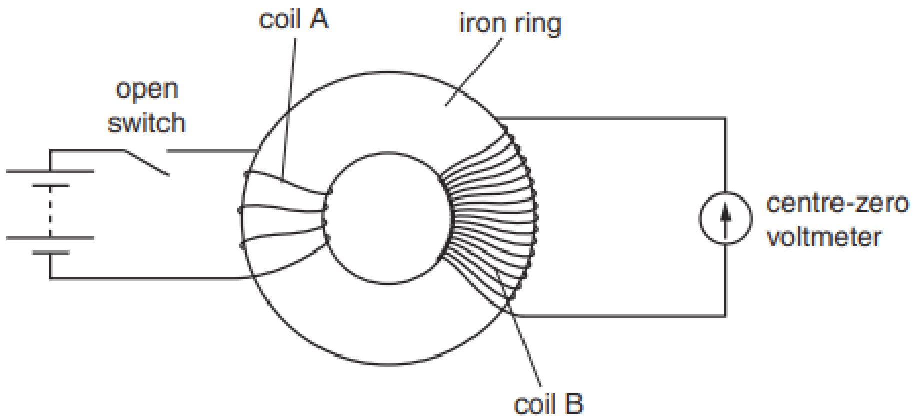


**Fig. 9.1**

The switch is closed. There is a current in coil A.

**(a)** On Fig. 9.1,

**(i)** mark the direction of the current in coil A, [1]

**(ii)** draw the magnetic field lines produced in the iron ring. [3]

**(b)** As the switch is closed, the voltmeter deflects to the right and then returns to zero.

**(i)** Explain why there is a deflection on the voltmeter.

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..........................................................................................................................

..........................................................................................................................[2]

**(ii)** The switch is opened. State and explain what happens to the deflection on the voltmeter.

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..........................................................................................................................

..........................................................................................................................[2]

**(iii)** Without changing coil A, state two changes to the apparatus that cause a greater deflection of the voltmeter.

1. ..........................................................................................................................

2. ..........................................................................................................................

[2]

**(c)** The battery in Fig. 9.1 is replaced by an alternating current (a.c.) supply. The output from coil B is used to power a lamp that is a long distance away. Each lead from coil B to the lamp has a resistance of 2.5 Ω. These leads are represented by the two resistors shown in Fig. 9.2.

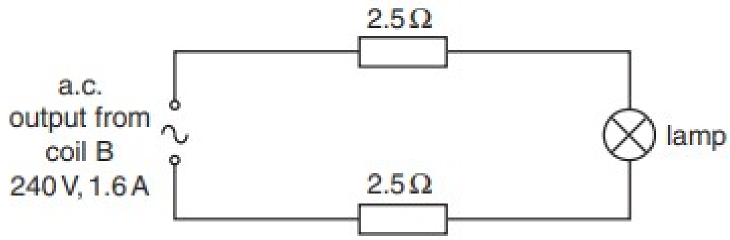


**Fig. 9.2**

The output voltage of coil B is 240 V and the current in the circuit is 1.6 A.

**(i)** Calculate the electrical power produced by coil B.

power = ................................................................[2]

**(ii)** Calculate the total power loss in the leads to the lamp.

power loss = ........................................................[3]

**46. M/J/12/21/Q8**

**(a)** A magnet is placed on a bench, as shown in Fig. 8.1a. End P of a rod is held above each end of the magnet in turn, as shown in Fig. 8.1b and in Fig. 8.1c. One end of the magnet is lifted off the bench in both cases.

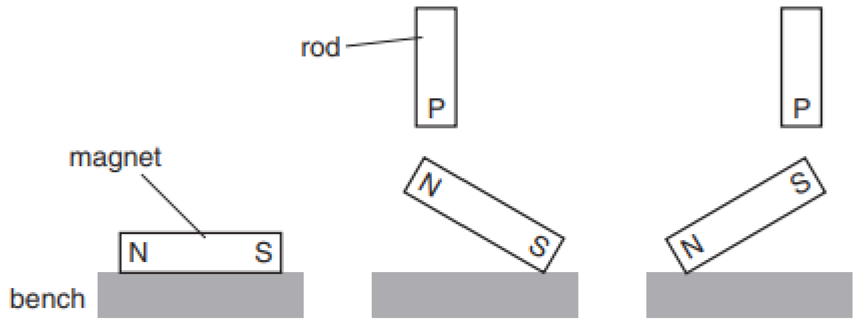


**Fig. 8.1a** **Fig. 8.1b** **Fig. 8.1c**

**(i)** Suggest what material the rod is made from.

............................................................................................................................[1]

**(ii)** Explain how the rod lifts each end of the magnet off the bench.

............................................................................................................................

............................................................................................................................

............................................................................................................................[2]

**(b)** Fig. 8.2 and Fig. 8.3 show views of a wire carrying a current downwards through a horizontal board.

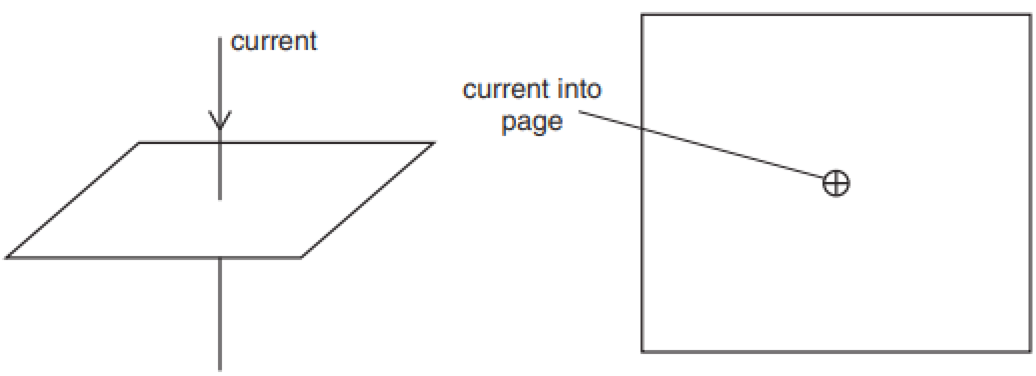


**Fig. 8.2** **Fig. 8.3** (viewed from above)

**(i)** On Fig. 8.3, draw the magnetic field due to the current in the wire. [2]

**(ii)** The magnetic field is stronger closer to the wire. State how the magnetic field lines indicate that the field is stronger.

............................................................................................................................

............................................................................................................................[1]

**47. O/N/11/22/Q5**

Fig. 5.1 shows part of an electric bell.

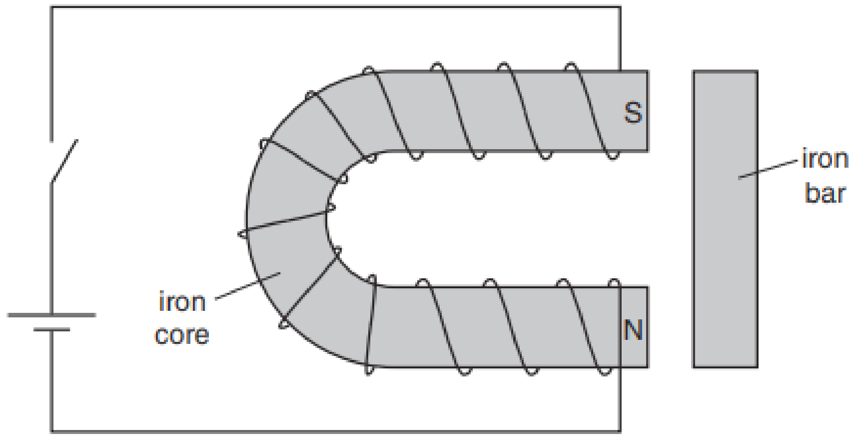


**Fig. 5.1**

A switch and a cell are in series with a length of wire coiled around an iron core.

The switch is closed and the current in the wire produces a south pole S and a north pole N at the ends of the core, as shown in Fig. 5.1. Magnetic poles are also produced in a small iron bar, placed near to the ends of the core.

**(a)** **(i)** On Fig. 5.1, mark with an N the position of the north pole produced in the iron bar and mark with an S the position of the south pole produced in the iron bar. [1]

**(ii)** State and explain what happens to the iron bar once it is magnetised.

..........................................................................................................................

..........................................................................................................................[2]

**(b)** The switch is opened and there is no current in the wire. State what happens to the magnetic poles in the iron bar.

..........................................................................................................................

..........................................................................................................................[1]

**48. O/N/11/21/Q6**

When a car is moving, its electrical equipment is powered by an a.c. generator.

**(a)** The coil of the a.c. generator is rotated by the car engine.

**(i)** On the axes in Fig. 6.1, sketch a graph of the output voltage of the coil against time for two rotations of the coil of the generator.


[1]

**Fig. 6.1**

**(ii)** The speed of the car increases and so does the speed of rotation of the coil.

State two changes that this causes to the output voltage.

change 1 ..........................................................................................................

..........................................................................................................

change 2 ..........................................................................................................

..........................................................................................................

[2]

**(b)** When the car engine is off, the current in a lamp from a 12V battery is 0.50A. Calculate the resistance of the lamp.

resistance = ......................................................[2]

**49. M/J/11/21/Q7**

Fig. 7.1 shows the structure of a circuit-breaker that uses an electromagnet. The circuit-breaker operates when the current is 10 A.

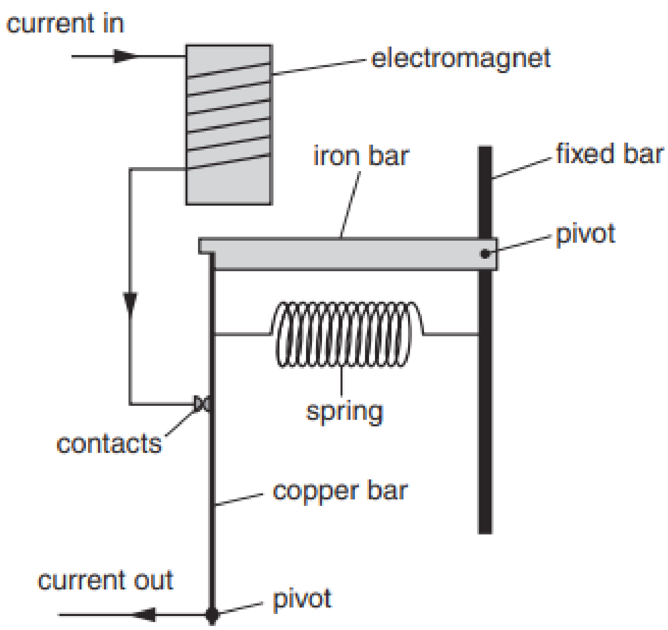


**Fig. 7.1**

**(a)** On Fig. 7.1, mark with an arrow the force on the iron bar caused by the electromagnet. [1]

**(b)** Suggest one reason why the iron bar does not move when the current is less than 10 A.

..........................................................................................................................................

.................................................................................................................................... [1]

**(c)** When the current is greater than 10 A, the circuit-breaker stops the current. Explain what happens in the circuit-breaker when this occurs.

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.................................................................................................................................... [3]

**(d)** State and explain how the electromagnet can be altered so that the circuit-breaker stops the current at less than 10 A.

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..........................................................................................................................................

.................................................................................................................................... [2]

**50. M/J/11/21/Q8**

**(a)** A wire carrying a current in a magnetic field experiences a force due to the current. On Fig. 8.1, insert the words **current**, **field** and **force** in the boxes to show the relative directions of the current, the magnetic field and the force.

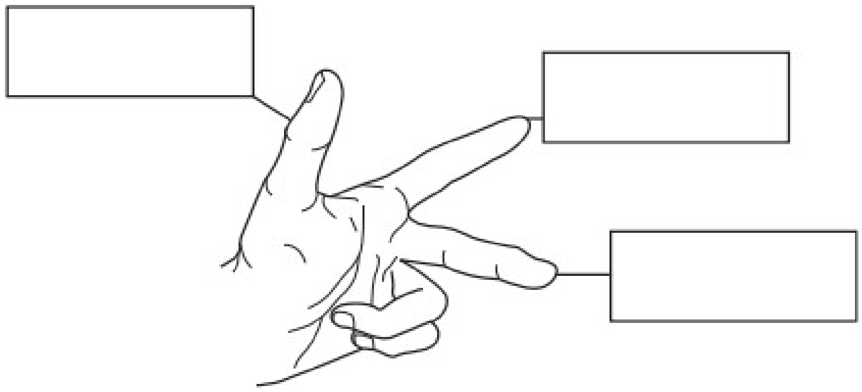

**Fig. 8.1**

[1]

**(b)** Fig. 8.2 shows a current-carrying coil ABCD in a magnetic field.

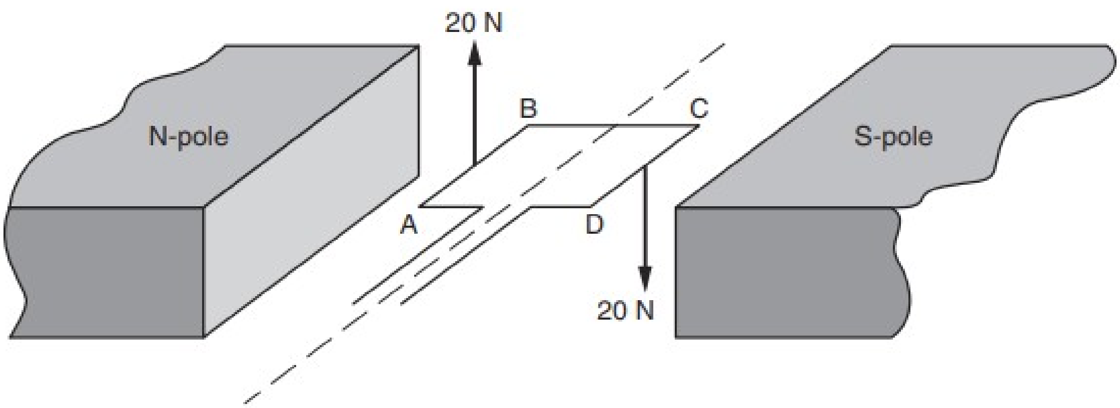


**Fig. 8.2**

Each side of the coil is 4.0 cm in length. The force on AB is 20 N and the force on CD is 20 N.

**(i)** Calculate the total moment caused by these forces.

moment = ........................................................... [2]

**(ii)** The moment is increased by using a stronger magnetic field.

State two other ways to increase the moment.

1. ..........................................................................................................................

2. ..........................................................................................................................

[2]

**51. M/J/11/21/Q9**

Fig. 9.1 shows a rotating magnet in an alternating current generator that is used to power a lamp.

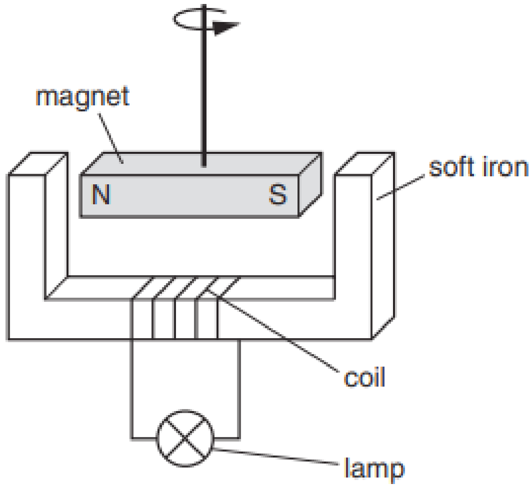


**Fig. 9.1**

**(a)** **(i)** State how an *alternating* current differs from a *direct* current.

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.................................................................................................................................... [1]

**(ii)** Explain, in detail, how **alternating** current is produced by the apparatus shown in Fig. 9.1.

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.................................................................................................................................... [4]

**(iii)** State two ways in which the current in the lamp may be increased.

1. ......................................................................................................................................

2. ......................................................................................................................................

[2]

**(b)** The generators at a power station produce a voltage of 25 000 V. This voltage is stepped up to 400 000 V by a transformer for long-distance transmission on overhead power lines. The voltage is later stepped down to 240 V.

**(i)** State and explain why the voltage is stepped up for long-distance transmission.

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.................................................................................................................................... [2]

**(ii)** Calculate the ratio of the number of turns in the primary coil of the step-up transformer to the number of turns in its secondary coil.

ratio = ........................................................ [1]

**(iii)** State one advantage and one disadvantage of using thicker wire in the overhead power lines.

advantage: ..........................................................................................................................

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disadvantage: ......................................................................................................................

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[2]

**(iv)** An electric drill of power 1000 W is used in a country where the mains voltage is 240 V.
State and explain the most appropriate fuse to use with this drill.
You should select a fuse from the following values: 1 A, 3 A, 4 A, 13 A.

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.................................................................................................................................... [3]

**52. O/N/10/22/Q6**

Fig. 6.1 shows a metal bar placed inside a vertical solenoid.

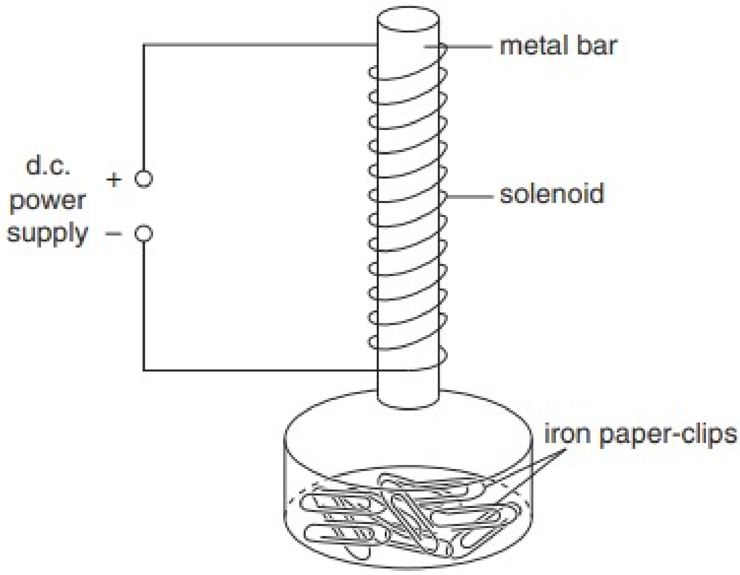


**Fig. 6.1**

The solenoid is a coil of several turns of insulated wire. A d.c. power supply is connected to the solenoid so that there is a current in it when the supply is switched on. The metal bar is a short distance above a small pile of iron paper-clips in a glass dish.

The power supply is

- switched on,
- left on for several seconds,
- then switched off.

Describe the behaviour of the paper-clips when this procedure is carried out using a metal bar of

**(a)** aluminium,

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..........................................................................................................................[1]

**(b)** iron,

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..........................................................................................................................[2]

**(c)** steel.

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..........................................................................................................................[2]

**53. O/N/10/21/Q6**

Fig. 6.1 shows the coil of a loudspeaker attached to a cardboard cone. One pole of a stationary cylindrical magnet lies near to the coil.

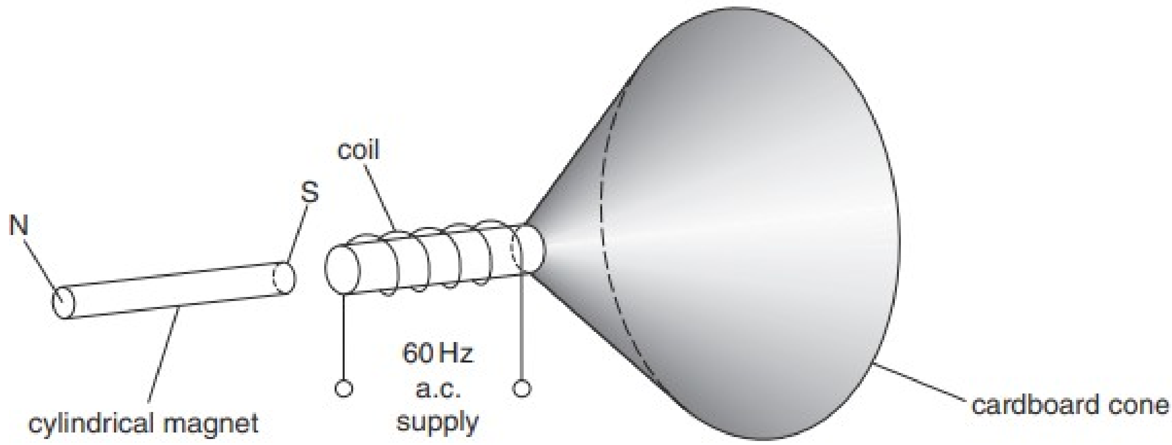


**Fig. 6.1**

There is an alternating current in the coil of the loudspeaker. A student hears the note produced.

**(a) (i)** Explain why the cone of the loudspeaker vibrates.

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.......................................................................................................................... [3]

**(ii)** Explain how the vibrating cone produces sound waves in the air.

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.......................................................................................................................... [2]

**(b)** A stronger cylindrical magnet is now used. State the difference in the note heard.

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.......................................................................................................................... [1]

**54. M/J/10/22/Q7**

Fig. 7.1 shows a compass.

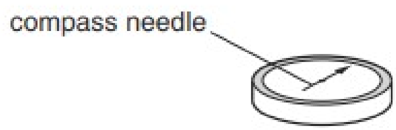


**Fig. 7.1**

The compass needle is a small magnet free to rotate. The head of the arrow on the compass needle is an N-pole.

**(a)** A bar magnet is placed between two compasses, as shown in Fig. 7.2.

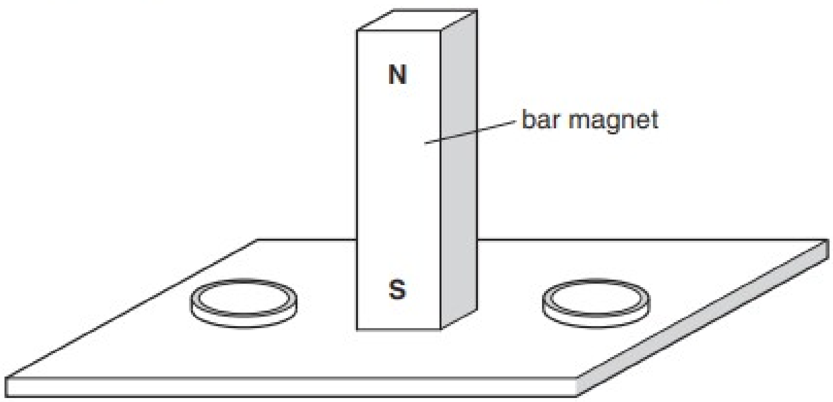


**Fig. 7.2**

On Fig. 7.2, draw the needles inside the two compasses and mark the N-pole of both compass needles. [2]

**(b)** Fig. 7.3 shows the structure of a relay.

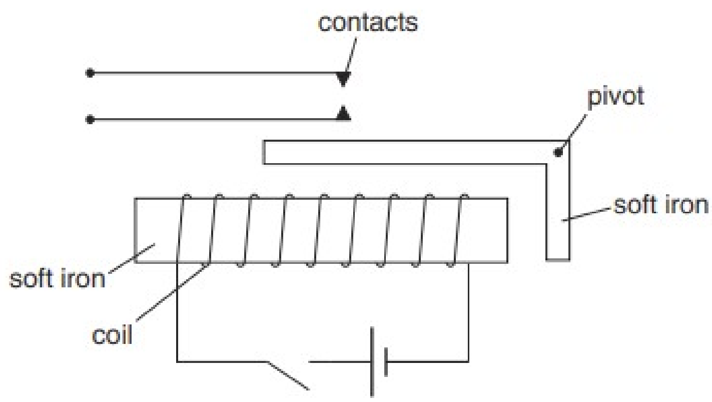


**Fig. 7.3**

Explain how closing the switch causes the contacts to close.

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.................................................................................................................................... [2]

**(c)** Fig. 7.4 is a circuit that includes a relay.

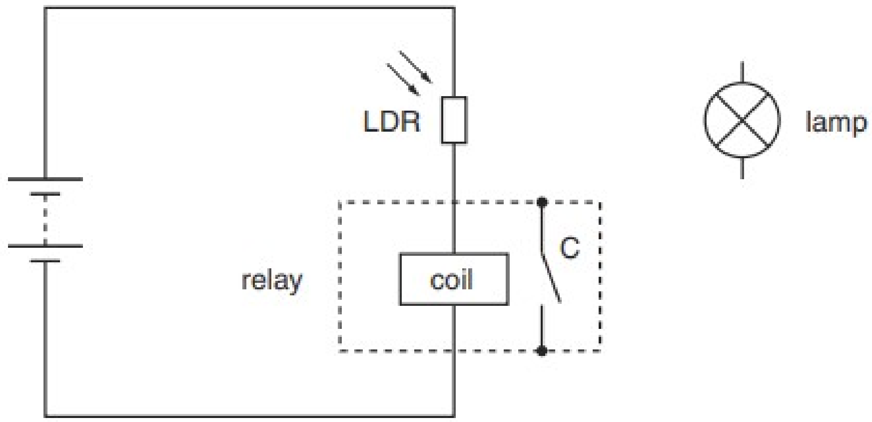


**Fig. 7.4**

When light shines on the light-dependent resistor (LDR), the relay contacts C close.

**(i)** State what happens to the resistance of the LDR when light falls on it.

............................................................................................................................. [1]

**(ii)** The circuit for the lamp, as shown in Fig. 7.4, is not complete.

On Fig. 7.4, draw the connections to the lamp, the contacts C and the battery that cause the lamp to switch on when light shines on the LDR. [2]

**55. M/J/10/21/Q8**

Fig. 8.1 shows two coils of wire wound on an iron ring. One coil is connected in series to a switch and a d.c. supply. The other coil is connected to a very sensitive centre-zero ammeter.

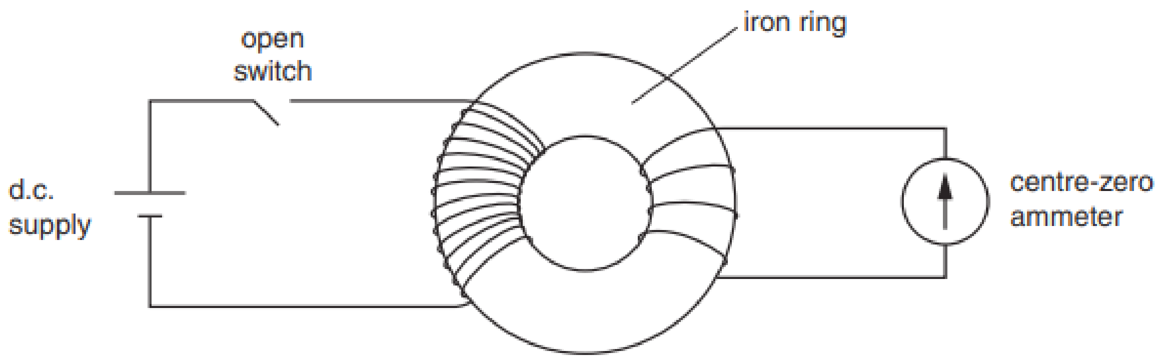


**Fig. 8.1**

At first the switch is open, as shown in Fig. 8.1.

The following actions are taken in turn.

Describe and explain what happens to the reading on the ammeter in each case.

**(a)** The switch is closed.

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..........................................................................................................................................[3]

**(b)** The switch is left closed for a long time.

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..........................................................................................................................................[1]

**(c)** The switch is opened.

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..........................................................................................................................................[2]

**56. M/J/09/Q7**

Fig. 7.1 shows two pieces of soft iron in the magnetic field of a strong permanent magnet.

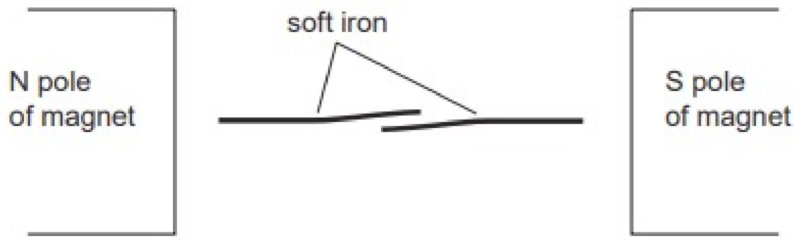


**Fig. 7.1**

The pieces of soft iron become magnetised.

**(a)** On Fig. 7.1, mark the magnetic poles produced at each end of both pieces of soft iron. [1]

**(b)** Fig. 7.2 shows a reed switch.

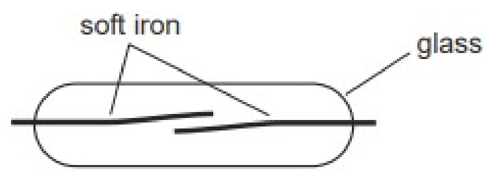


**Fig. 7.2**

The reed switch is placed between the poles of the strong permanent magnet.
State and explain what happens.

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.................................................................................................................................... [2]

**(c)** Fig. 7.3 shows two separate electrical circuits.

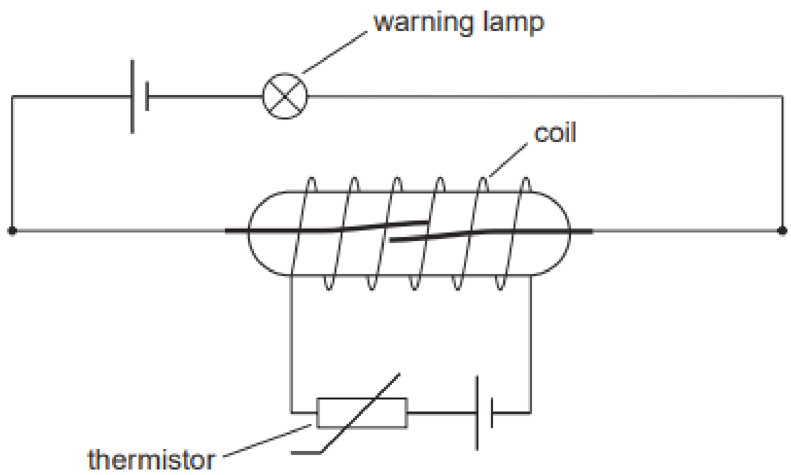


**Fig. 7.3**

One circuit consists of a reed switch, a cell and a warning lamp. The other circuit consists of a thermistor, another cell, and a coil wound round the reed switch. The thermistor is at the same temperature as the air around it.

**(i)** State what happens to the thermistor when the temperature of the air rises.

............................................................................................................................ [1]

**(ii)** Explain why the warning lamp lights up when the air temperature rises.

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............................................................................................................................

............................................................................................................................ [2]

**57. M/J/08/Q7**

**(a)** Fig. 7.1 shows a straight wire between the poles of a magnet. The wire carries a current into the page.

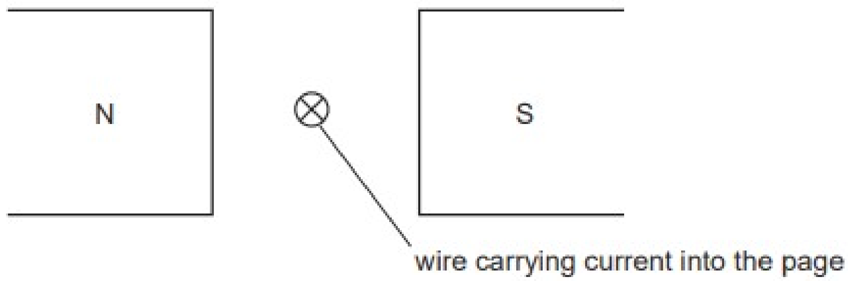


**Fig. 7.1**

**(i)** State the direction of the magnetic field between the poles of the magnet.

.......................................................................................................................... [1]

**(ii)** On Fig. 7.1, draw an arrow to show the direction of the force acting on the wire. [1]

**(b)** Fig. 7.2 shows two wires.
Each wire carries a current into the page.

**Fig. 7.2**

**(i)** On Fig. 7.2, draw the magnetic field due to the currents in the wires. [3]

**(ii)** There is a force on each wire due to the current in the other wire.

On Fig. 7.2, draw an arrow on each wire to show these forces. [1]

## 58. O/N/07/Q7

Fig. 7.1 shows apparatus that can be used to make an electromagnet or a permanent magnet.

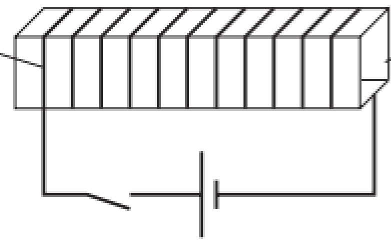


**Fig. 7.1**

Four rods are available. They are made of aluminium, soft iron, steel and wood.

**(a)** **(i)** State which rod is used to make a permanent magnet.

.............................................................................................................. [1]

**(ii)** Describe how the apparatus is used to make a permanent magnet.

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.............................................................................................................. [1]

**(b)** A computer component is screened from external magnetic fields by placing it in a box, as shown in Fig. 7.2.

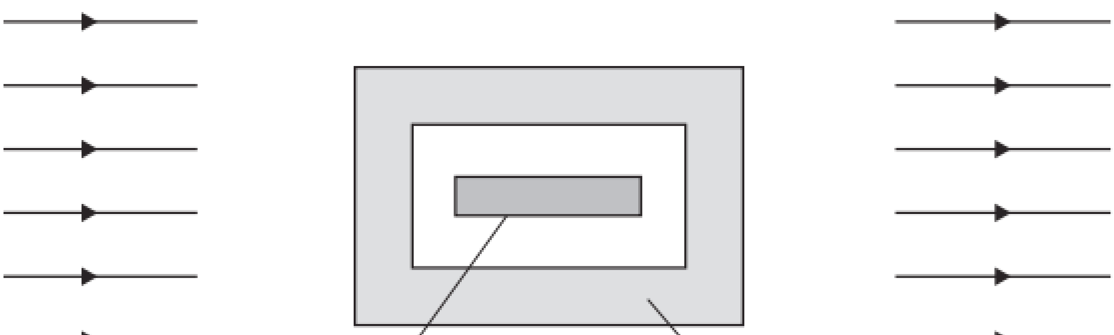


**Fig. 7.2**

There is a strong magnetic field outside the box. The magnetic field lines have not been drawn near the box.

**(i)** State the best choice for the material of the box.

.............................................................................................................. [1]

**(ii)** On Fig. 7.2, join the magnetic field lines on the left of the box to those on the right, showing the pattern of the magnetic field. [2]

**59. M/J/07/Q6**

Fig. 6.1 shows a coil of wire wound on a cardboard tube.

There is a d.c. current in the coil. The direction of the current is shown in the key.

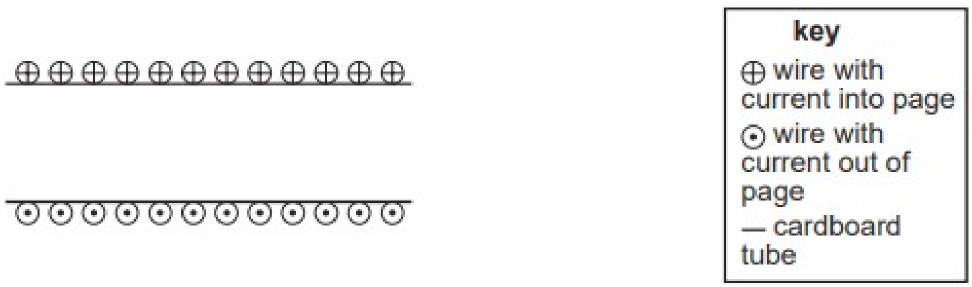


**Fig. 6.1**

**(a)** On Fig. 6.1, draw the magnetic field produced by the coil. [3]

**(b)** Fig. 6.2 shows a simple loudspeaker that uses the coil shown in Fig. 6.1 attached to a paper cone.

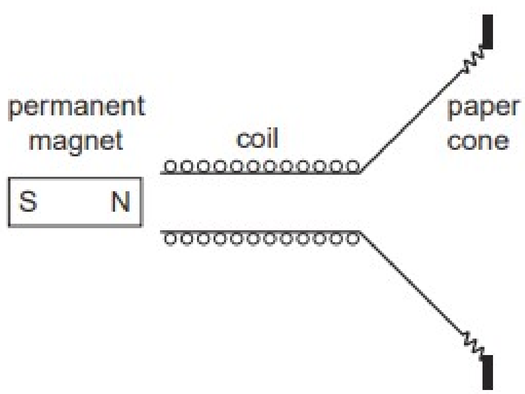


**Fig. 6.2**

The coil is connected to a signal generator.

There is an alternating current of frequency 100 Hz in the coil.

**(i)** State what is meant by a *frequency of 100 Hz*.

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.......................................................................................................................... [1]

**(ii)** Describe and explain the movement of the coil.

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.................................................................................................................................. [3]

**60. O/N/06/Q7**

Fig. 7.1 shows one way to demonstrate an electrical effect.

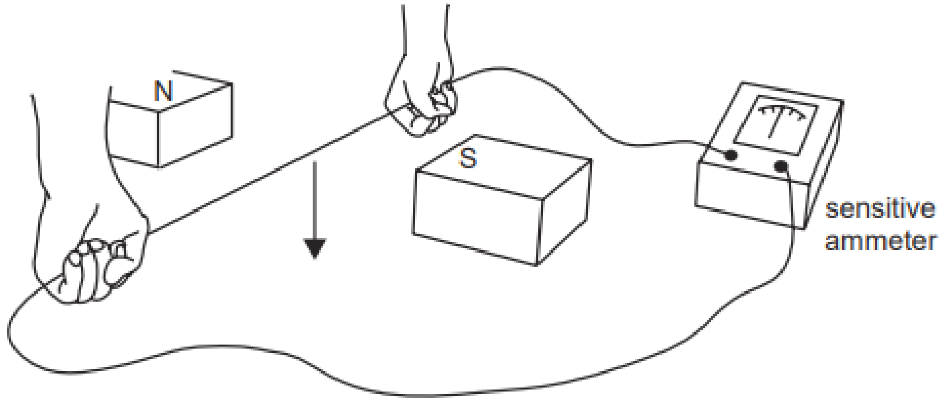


**Fig. 7.1**

As the wire is moved downwards between the magnetic poles, the needle of the ammeter deflects to the right.

**(a)** State the name of this electrical effect.

.......................................................................................................................... [1]

**(b)** State what happens to the needle of the ammeter when the wire is moved upwards between the magnetic poles.

..........................................................................................................................

.......................................................................................................................... [1]

**(c)** State and explain what happens when the wire is held stationary between the magnetic poles.

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.......................................................................................................................... [2]

**61. M/J/06/Q3**

Fig. 3.1 shows the construction of a simple a.c. generator. When the coil is rotated an e.m.f. is induced in the coil.

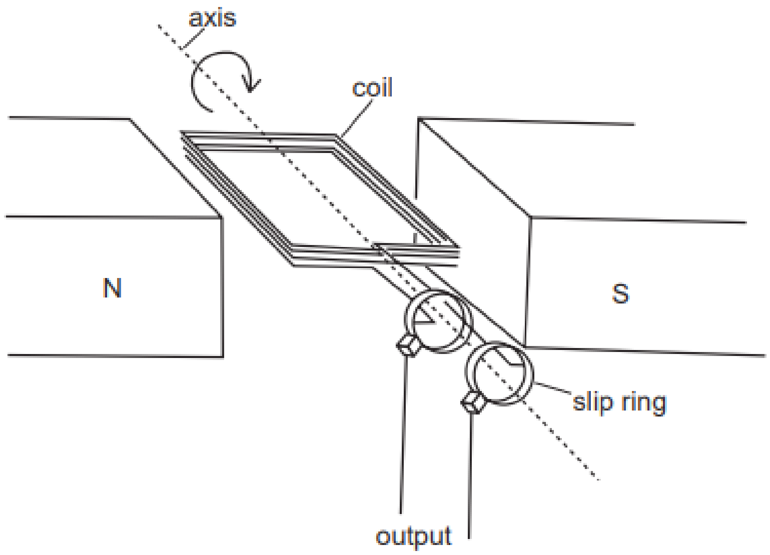


**Fig. 3.1**

**(a)** Explain why an e.m.f. is induced.

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..........................................................................................................................................

.................................................................................................................................. [2]

**(b)** State the purpose of the slip rings.

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.................................................................................................................................. [1]

**(c)** The direction of the current in the coil can be found from Lenz's law.

State Lenz's law.

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.................................................................................................................................. [1]

**(d)** The induced e.m.f. can be increased by rotating the coil faster. State one other way in which the e.m.f. can be increased.

..........................................................................................................................................

.................................................................................................................................. [1]

**62. M/J/06/Q7**

Fig. 7.1 shows an electrical circuit and a cathode-ray oscilloscope (C.R.O.).

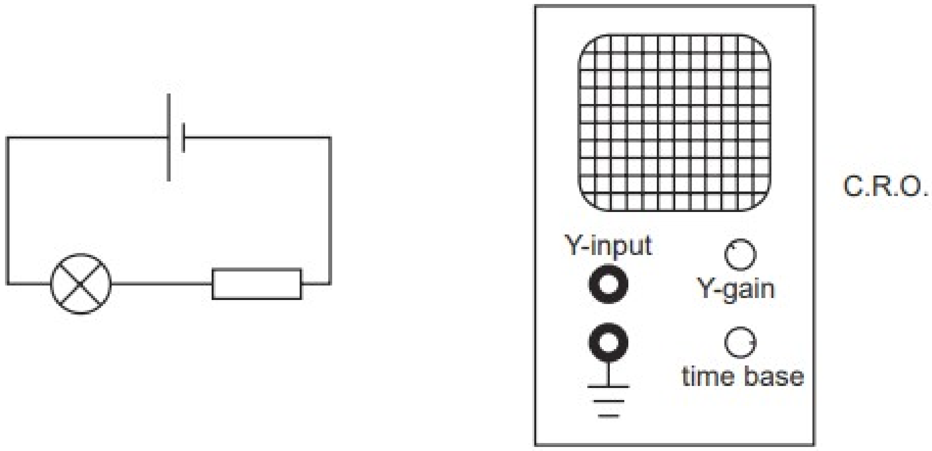


**Fig. 7.1**

**(a)** On Fig. 7.1, draw the connections you would make to enable the C.R.O. to measure the potential difference (p.d.) across the resistor. [1]

**(b)** Fig. 7.2 shows the trace on the screen before and after the connections are made.

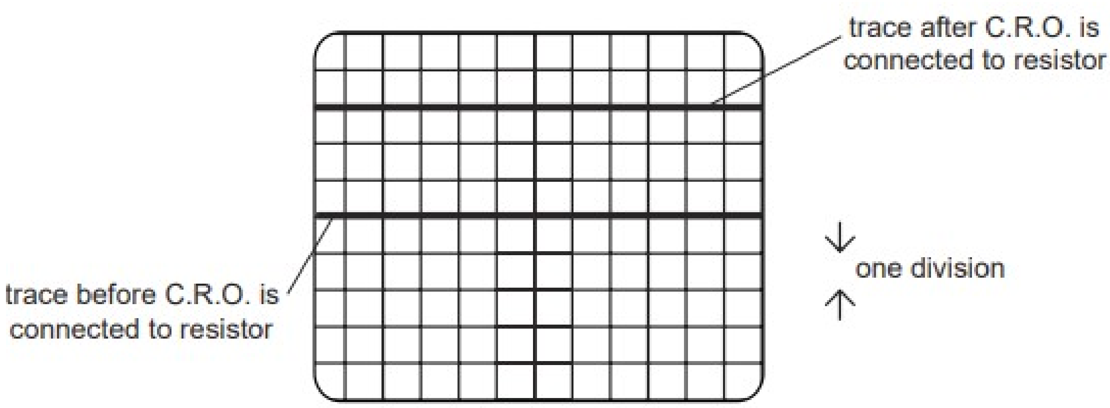


**Fig. 7.2**

The Y-gain control is set at 2 V for each division on the screen.

**(i)** State the value of the p.d. across the resistor.

p.d. = .............................................................. [2]

**(ii)** The Y-gain control is altered to 4 V for each division.

On Fig. 7.2, draw the new trace seen on the screen. [1]

**63. O/N/05/Q7**

Fig. 7.1 shows a coil of wire wound around a rectangular tube.

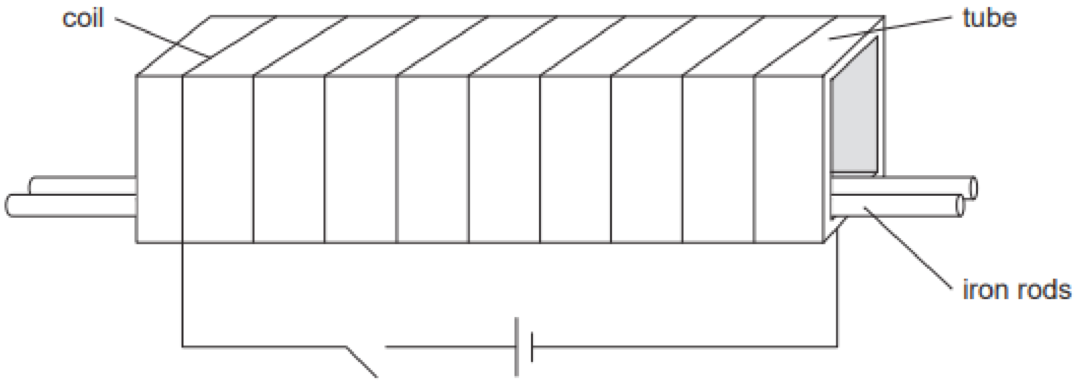


**Fig. 7.1**

**(a)** Two iron rods are placed next to each other at the bottom of the tube. When the current is switched on, the two rods repel each other. They move to the sides of the tube.

Explain why the two iron rods repel.

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.................................................................................................................................... [2]

**(b)** An iron rod and a similar copper rod are placed next to each other at the bottom of the tube. State and explain what, if anything, happens to the rods when the current is switched on.

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..........................................................................................................................................

.................................................................................................................................... [2]

**64. M/J/05/Q5**

Fig. 5.1 shows a coil of wire wrapped around a plastic tube. Inside the tube are two pieces of soft iron. When the switch is closed, the compass needles point in the direction of the magnetic field produced at each position. You may ignore the magnetic field of the Earth in this question.

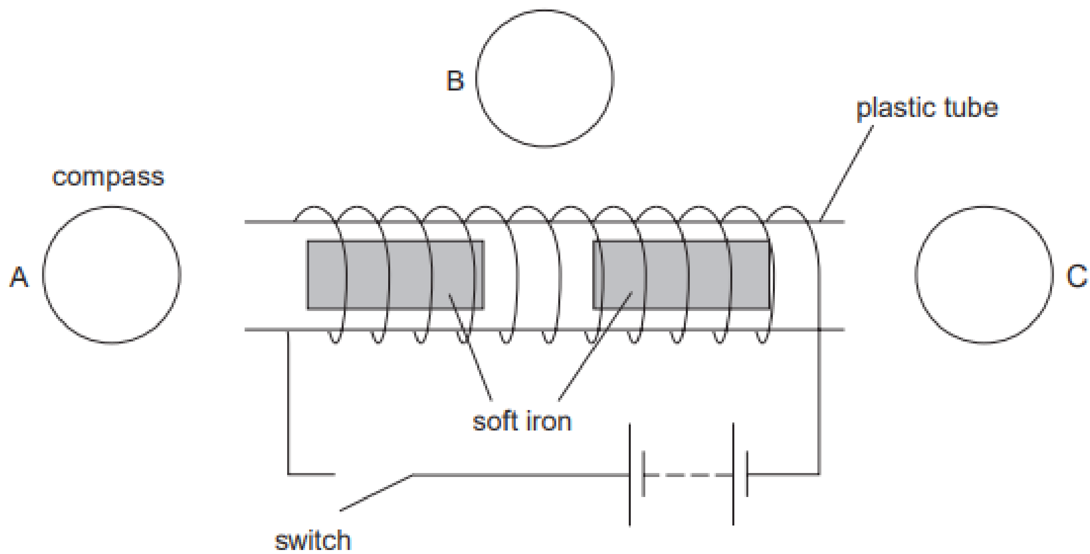


**Fig. 5.1**

**(a)** On Fig. 5.1 mark arrows, in compasses A, B and C, to show the direction of the magnetic field at each position after the switch has been closed. [2]

**(b)** When the switch is closed, the two pieces of soft iron in the tube become magnets and move.

**(i)** On Fig. 5.1, mark the poles formed on each piece of soft iron. [1]

**(ii)** State and explain how the pieces of iron move.

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..........................................................................................................................

.................................................................................................................... [2]

**(c)** State the effect on the magnetic field of

**(i)** reversing the direction of the current,

..........................................................................................................................

.................................................................................................................... [1]

**(ii)** reducing the size of the current.

..........................................................................................................................

.................................................................................................................... [1]

**65. O/N/04/Q7**

Fig. 7.1 shows high voltage cables used to transmit electrical energy.

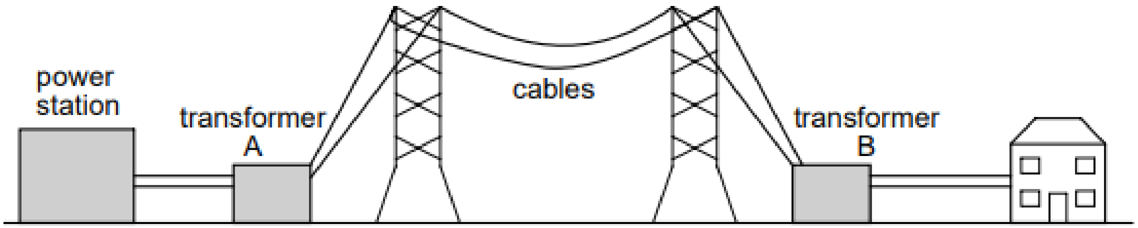


**Fig. 7.1**

**(a)** State the purpose of transformer B.

..............................................................................................................................[1]

**(b)** In the space below, draw a labelled diagram to show the structure of transformer B.

[3]

**(c)** **(i)** Explain why high voltages are used to transmit electrical power.

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**(ii)** Fig. 7.2 shows how the loss of thermal energy from a cable varies with the thickness of the cable.

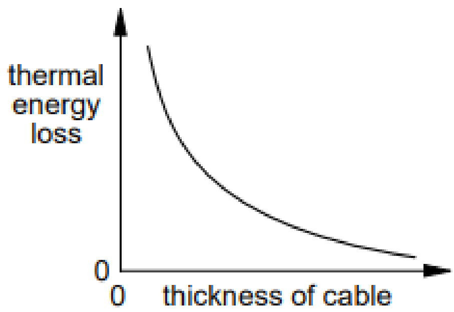


**Fig. 7.2**

Explain why the loss of thermal energy is less if the cable is thicker.

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..........................................................................................................................................
[4]

**66. M/J/04/Q5**

Fig. 5.1 shows a piece of recording tape passing under the recording head of a tape recorder. An alternating current is passed through the coil. The tape is coated with a magnetic material that becomes magnetised.

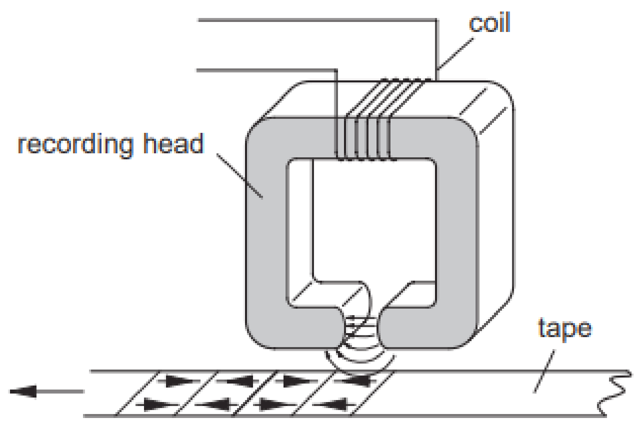


**Fig. 5.1**

**(a)** **(i)** Explain why the tape becomes magnetized.

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**(ii)** Fig. 5.1 shows that sections of the tape are magnetised in opposite directions. Explain why they become magnetised in opposite directions.

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**(iii)** The tape is moved faster past the recording head. State how this changes the pattern on the tape.

..........................................................................................................................

..........................................................................................................................

[3]

**(b)** **(i)** Explain why the coating on the tape must be of a permanent magnetic material.

..........................................................................................................................

..........................................................................................................................

**(ii)** State the name of a permanent magnetic material.

..........................................................................................................................

[2]